
Opinion Dynamics From A Game Theoretic Perspective

Balaji R

B.Tech Mathematics And Computing
IISc Bangalore

Abstract

In a social network, where people's opinions on a topic are constantly evolving due to people's interaction with each other, the authors model this as a non linear dynamical system where people's preferences are evolving based on the influence of others. Each agent looks to maximize his or her utility which depends on a constant self-preference, social influence, and includes a resource penalty for bounding the opinion. Based on this model, the authors analyze the asymptotic behavior, consensus and Nash equilibria in various special cases where there are no antagonistic or negative influential relations. The authors also analyze the egalitarian and utilitarian price of anarchy under the same assumption as a measure of how the equilibrium profile satisfies all agents and the network as a whole. The work looks to explore the impact of bounded resource constraints under the assumption that agents are myopically looking to maximize their utilities. Finally, under the paper's model, the oscillatory behavior of a two-agent social network is studied. The **experiments** conducted in the paper were reproduced and **theoretical results verified**. Further, I have **extended** the work to model multiple topics. Additionally, I have given a model for the situation where one has multiple internal preferences and is resource-constrained to optimize the preference term.

1 Introduction

Opinion dynamics deals with the study of how beliefs evolve with time, and spread within social groups or networks due to influence of connections in the networks. It has far-reaching implications in diverse sectors such as policy, public health, sociology, finance or economics.

The study by Wankhede et al[1] builds upon the existing game theoretic idea of dynamic, myopic utility maximization, and studies the impact of resource constraints. The scenario in the paper is agents have different opinions on how much resources to allocate to tasks. Each agent has heterogeneous amount of resources which limits his or her opinion. This is captured by a resource penalty term in the dynamics.

Section 2 is a literature review giving background on existing work in opinion dynamics. In section 2.1, we study classical models for opinion dynamics building up from the French-DeGroot model[2,3] and an interpretation of social power or the ability to be influential over a social network. Then we introduce the continuous time analogue, the Abelson model[4]. Next, we introduce stubbornness in agents biased towards an internal preference in the Taylor[5] and Friedkin-Johnson[6] models. Next we introduce the concept of homophily via the Hegselmann-Krause model where each agent is influenced by people who have opinions close to hers. This is also a bounded confidence model[10] where in each agent possesses and acts using bounded rationality or resources. Next we consider the effect of negative influence and the boomerang effect through Altafini[8]'s model which introduces balance theory. Finally, we look at the DeGroot Friedkin model[20], where the ability to influence is quantified and incorporated into the social network.

In section 2.2, we look at game theoretic models and analyze the work of different papers. We start by looking at cooperative control problems[22] such as consensus and how we can use potential games, and their extension into weakly acyclic games are a good way to tackle such problems. Next we look at papers that view opinion dynamics as a best response dynamics to utility maximization problems. An analysis of the behavior and consequences of stubborn agents in a social network[23], convergence rates and equilibria is carried out. Next, we study Kleinberg's paper[24] in FOCS where the price of anarchy or the distance from global optimality of stubborn agents in the discrete time dynamics is analysed. We analyse the work of Etasami et al[27] for a game theoretic view of the Hegselmann-Krause model, where synchronous and asynchronous updates are introduced, and the latter being shown to be the best response dynamics to a "network formation" game. The work of Park et al[30], that discusses cooperative behaviour in repeated games model under a non-linear opinion dynamics framework is studied. It was seen that successful strategies were nice, forgiving, provokable, and clear. Next, we study the game of dynamic influence maximization in a paper by Etasami et al[26]. Here, influencers hold advertising campaigns with limited and try to boost players' opinions which is modeled as a non-linear dynamical system with jumps due to campaigns. Influencers solve an optimization problem for efficient budget allocation. Finally, we look at the effect of conformity, where people tend to move their opinion closer to the public average and manipulation, where people give false ratings to bring the global average closer to their true opinion in a paper by Etasami et al[31].

In section 3, we study the work of Wankhede et al which analyses the impact of the resource penalty term in the opinion dynamics. In section 4, we reproduce most of the experiments and appreciate the theoretical results by verifying them through simulations.

Finally, in appendix A, the work has been extended to consider opinions on multiple topics, where each agent has an opinion vector, and a way to incorporate resource constraint to add the coupling effect into opinions of different topics. Additionally in appendix B, we also look at the case when agents can have multiple preferences about a topic and is simultaneously resource constrained. Thus the agent partly optimises the gradient of the preference part of the gradient and partly just follows the crowd.

2 Literature Review

2.1 Classic Models For Opinion Dynamics

2.1.1 French-DeGroot Model[2,3]

The French-DeGroot model describes how opinions evolve within a network of n agents over discrete time steps. Each agent i maintains a scalar opinion $x_i \in \mathbb{R}$. The model is governed by an influence matrix $W = (w_{ij})$, where each entry $w_{ij} \geq 0$ represents the weight assigned by agent i to agent j . These influence weights determine how opinions are updated, with higher values indicating stronger influence.

The opinion dynamics follow the iterative update rule:

$$x(k+1) = Wx(k), \quad k = 0, 1, 2, \dots \quad (1)$$

where $x(k) = (x_1(k), \dots, x_n(k))^T$ denotes the opinion vector at time step k . This equation captures how each agent's opinion is shaped by the weighted combination of others' opinions.

In this framework, the weight w_{ij} represents the contribution of agent j 's opinion to the opinion of agent i at the next time step. The self-influence weight $w_{ii} \geq 0$ indicates the extent to which an agent incorporates external opinions. If $w_{ii} = 0$, the agent is fully open to influence from others, whereas if $w_{ii} = 1$ with $w_{ij} = 0$ for all $j \neq i$, the agent remains steadfast in its initial opinion, acting as a stubborn agent.

More generally, opinions can be represented as vectors of dimension m . Each agent's opinion is then given by a row vector $x_i = (x_{i1}, \dots, x_{im})$. Stacking these row vectors together forms the opinion matrix $X = (x_{il}) \in \mathbb{R}^{n \times m}$. The update rule extends naturally to the matrix form:

$$X(k+1) = WX(k), \quad k = 0, 1, 2, \dots \quad (2)$$

Each column $x_i(k) = (x_{1i}(k), \dots, x_{ni}(k))^T \in \mathbb{R}^n$ of $X(k)$ follows the same evolution rule as in the scalar case. The model described by this matrix equation with a general stochastic matrix W , i.e. with row sum 1, is referred to as the French-DeGroot model.

French initially proposed several conditions for reaching a consensus, defined as the convergence

$$x_i(k) \xrightarrow{k \rightarrow \infty} x^* \quad (3)$$

where all opinions eventually align to a common value.

Convergence and Regularity:

A model is said to be *convergent* if, for any given initial opinion vector $x(0)$, the sequence of opinions stabilizes, meaning that the limit

$$x(\infty) = \lim_{k \rightarrow \infty} x(k) = \lim_{k \rightarrow \infty} W^k x(0) \quad (4)$$

exists. A convergent model achieves consensus if all agents' opinions ultimately converge to the same value, i.e.,

$$x_1(\infty) = x_2(\infty) = \dots = x_n(\infty), \quad (5)$$

regardless of the initial condition $x(0)$. The concepts of convergence and consensus are closely related to the properties of the influence matrix W .

A stochastic matrix W is considered *regular* if the limit

$$W^\infty = \lim_{k \rightarrow \infty} W^k \quad (6)$$

exists. Furthermore, W is said to be *fully regular* if all rows of W^∞ are identical, meaning that for some vector $p_\infty \in \mathbb{R}^n$, the authors have

$$W^\infty = \mathbf{1} p_\infty^T. \quad (7)$$

This ensures that the influence of all agents ultimately aligns towards a common consensus.

Some Results:

Theorem 2.1. *For an irreducible stochastic matrix W , the model is convergent if and only if W is primitive, i.e., W^k is a positive matrix for sufficiently large k . In this case, consensus is also achieved.*

Theorem 2.2. *If the agents' self-weights satisfy $w_{ii} > 0$ for all i , then the model is convergent.*

French's primary interest was to determine a quantitative measure of an agent's social power, which represents its ability to influence the collective opinion x^* . This social power is mathematically quantified as the agent's weight in the consensus opinion. Specifically, when consensus is reached, the final opinion can be expressed as a weighted sum of the initial opinions:

$$x^* = \sum_{i=1}^n \pi_i x_i(0), \quad (8)$$

where the weights π_i sum to 1 and indicate the long-term influence of each agent.

2.1.2 Social Power In The French Degroot Model

In the French-DeGroot model, the concept of social power refers to an agent's ability to influence the group's collective opinion. This is quantified by examining how each agent's initial opinion contributes to the group's final consensus.

Mathematically, if the model reaches a consensus, the final opinion can be represented as a weighted sum of the agents' initial opinions. The weight associated with each agent's initial opinion in this sum defines their social power.

Social power is inherently linked to the concept of a stationary distribution in a Markov chain. Essentially, if the authors interpret the influence matrix W as the transition matrix of a Markov process, the long-term behavior of the system is described by the stationary distribution of this Markov chain.

Each individual's influence, or social power, corresponds to how much their initial opinion contributes to the group's final consensus. In other words, if a person has a high weight in this stationary distribution, it means their opinion holds more influence in shaping the collective belief over time.

This provides a natural way to measure influence within a social network—agents with higher stationary probabilities are those who exert stronger long-term effects on the group's opinion.

2.1.3 Stubborn Agents in the French-DeGroot Model

Although consensus is a typical behavior of the model, there are situations where opinions do not converge to a single consensus but instead split into multiple clusters. One of the key reasons for this phenomenon is the presence of stubborn agents.

Definition 1. (Stubbornness) *An agent is said to be stubborn if its opinion remains unchanged, regardless of the influence of other agents.*

2.1.4 Abelson Model[4]

In his influential work, Abelson introduced the continuous time counterpart of the French-DeGroot model.

The authors can write the continuous analogue of the FD model as:

$$\dot{x}_i(t) = \sum_{j \neq i} a_{ij}(x_j(t) - x_i(t)), \quad i = 1, \dots, n. \quad (9)$$

Let $A = (a_{ij})$ be a non-negative (but not necessarily stochastic) matrix of infinitesimal influence weights.

The linear Abelson model, has the matrix form:

$$\dot{x}(t) = -L[A]x(t), \quad (10)$$

where $L[A]$ is the Laplacian matrix of A .

This model can always be shown to converge unlike the French-DeGroot model.

2.1.5 Friedkin-Johnson Model[6]

First, the authors introduce prejudices or self preferences into the Abelson model, tweaking the structure as follows:

$$\dot{x}_i(t) = \sum_{j=1}^n a_{ij}(x_j(t) - x_i(t)) + \sum_{k=1}^m b_{ik}(s_k - x_i(t)). \quad (11)$$

The above model, known as the Taylor model[5], involves n agents with opinions $x_1, \dots, x_n \in \mathbb{R}$ and $m \geq 1$ communication sources (such as mass media), providing static opinions $s_1, \dots, s_m \in \mathbb{R}$. This model involves the non-square $n \times m$ nonnegative matrix $B = (b_{ik})$ of "persuasibility constants," which describe the influence of the communication/external sources on the agents that create internal self-preferences.

Another transformation allows reducing this to a less general model, where each agent may have its own "communication source" or prejudice:

$$\dot{x}_i(t) = \sum_{j=1}^n a_{ij}(x_j(t) - x_i(t)) + \gamma_i(u_i - x_i(t)), \quad (12)$$

where $\gamma_i \geq 0$.

Definition 2. (Prejudiced Agents) *Given a group of n agents governed by the Taylor model, the authors call agent i prejudiced if $\gamma_i > 0$; the external inputs u_i are referred to as the prejudices of the corresponding agents.*

The Friedkin-Johnson model is the discrete-time counterpart of the Taylor model, with dynamics given by:

$$x(k+1) = \Lambda W x(k) + (I - \Lambda)u. \quad (13)$$

Similar to DeGroot's dynamics, the Friedkin-Johnsen model employs a stochastic matrix of social influences W . Besides this matrix, a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ is introduced, where $\lambda_i \in [0, 1]$ stands for the susceptibility of agent i to the process of social influence.

Here u is a constant vector of the agents' prejudices. The susceptibilities' complements $1 - \lambda_i$ play the same role as the coefficients γ_i in Taylor's model.

Unlike many models of opinion formation, proposed in the literature, this model has been experimentally validated for small and medium-size groups.

2.1.6 Time Varying Extensions[11]

The authors start with a time-varying counterpart of the French-DeGroot model, where W is replaced by a sequence $(W(k))_{k \geq 0}$:

$$x(k+1) = W(k)x(k), \quad k = 0, 1, \dots \quad (14)$$

The following lemma gives sufficient conditions of convergence of opinions:

Lemma 2.3. *Let $\delta > 0$ exist such that the sequence of $n \times n$ stochastic matrices $(W(k))_{k \geq 0}$ satisfies at any time $k \geq 0$ the following three conditions:*

- (a) (Non-vanishing couplings) $w_{ij}(k) \in \{0\} \cup [\delta, 1]$;
- (b) (Self-confidence) $w_{ii}(k) \geq \delta$ for all $i \in \{1, \dots, n\}$;
- (c) (Type-symmetry) $w_{ij}(k) > 0 \iff w_{ji}(k) > 0$.

Then the limit $\bar{x} = \lim_{k \rightarrow \infty} x(k)$ exists for any $x(0) \in \mathbb{R}^n$, being an equilibrium point: $W(k)\bar{x} = \bar{x}$ for sufficiently large k . If agents i and j interact persistently,

$$\sum_{k=0}^{\infty} w_{ij}(k) = \infty, \quad (15)$$

then their limit opinions coincide, $\bar{x}_i = \bar{x}_j$.

Similarly, one can extend Abelson's model:

$$\dot{x}(t) = -L[A(t)]x(t), \quad \forall t \geq 0 \quad (16)$$

The convergence criteria are similar to the above lemma, involving type symmetry.

2.1.7 Bounded confidence models[10]

Like-minded individuals interact more often and intensively than dissimilar people. Humans interact much more intensively with similar people and accept dissimilar opinions with discretion. This behavior is referred to as "homophily".

The variability of the "contact rates" $a_{ij}(t)$ in the time-varying Abelson model can express that people tend to locomote into groups that share their attitudes and out of groups that do not agree with them.

Bounded confidence models stipulate that individuals are totally insensitive to opinions, falling outside their confidence sets.

One of them, the Hegselman-Krause model has become widely known after being published.

2.1.8 Hegselmann-Krause(HK) Model[7]

The HK model deals with n agents, whose real opinions $x_i \in \mathbb{R}$ constitute the opinion vector $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. It introduces the fixed range of confidence $d > 0$ and defines the closed set $[x_i - d, x_i + d] \subset \mathbb{R}$ as the confidence interval of agent i . Each agent i ignores the opinions beyond their confidence interval, interacting only with a group of trusted individuals:

$$I_i(x) = \{j : |x_j - x_i| \leq d\} \ni i. \quad (17)$$

The opinion of agent i evolves at each step according to:

$$x_i(k+1) = \frac{1}{|I_i(x(k))|} \sum_{j \in I_i(x(k))} x_j(k), \quad i \in \{1, \dots, n\}. \quad (18)$$

For multidimensional opinions $x_i(k) \in \mathbb{R}^m$, the authors choose some norm $\|\cdot\|$ on $\mathbb{R}^m$, and the confidence interval for agent i is replaced by the ball:

$$\{\xi \in \mathbb{R}^m : \|\xi - x_i\| \leq d\}. \quad (19)$$

Hence, the set of trusted individuals is defined as:

$$I_i(x) = \{j : \|x_j - x_i\| \leq d\}. \quad (20)$$

The direct extension of the scalar HK model into continuous time is:

$$\dot{x}_i(t) = \sum_{j: |x_j(t) - x_i(t)| < d} (x_j(t) - x_i(t)), \quad i \in \mathbb{R}. \quad (21)$$

To avoid numerical instabilities caused by discontinuities, the following smoothed modification of the HK model can be introduced:

$$\dot{x}_i(t) = \sum_j s(x_j - x_i)(x_j(t) - x_i(t)), \quad x_i(t) \in \mathbb{R}, \quad (22)$$

where $s : \mathbb{R} \rightarrow \mathbb{R}^+$ stands for some even continuous function.

The following theorem talks about the asymptotic behavior of the HK model:

Theorem 2.4. *For any initial condition $x(0)$, the HK dynamics terminates in a finite number of steps:*

$$x(k) \equiv \bar{x}, \quad \forall k \geq k^*, \quad (23)$$

where the final opinion $\bar{x}$ and the termination time depend on $x(0)$ and d . After the model's termination, any agents i, j either are in consensus $\bar{x}_i = \bar{x}_j$ or distrust each other $|\bar{x}_i - \bar{x}_j| > d$. Furthermore, the HK model with n agents terminates in no more than $O(n^3)$ steps.

2.1.9 Adding stubbornness to HK

The dynamics of the French-DeGroot model change significantly in the presence of stubborn individuals, who maintain their opinions unchanged. A further relaxation of the stubbornness assumption leads to the Friedkin-Johnsen (FJ) model, where some agents exhibit partial stubbornness (prejudice). These agents incorporate others' opinions while remaining anchored to their initial opinions, factoring them into each step of the opinion iteration. Similar extensions have also been proposed for the HK model.

The proposed algorithm by Hegselmann and Krause to incorporate stubbornness is as follows:

Consider n agents with m -dimensional opinions $x_1, \dots, x_n \in \mathbb{R}^m$ and fix one point $T \in \mathbb{R}^m$ in the opinion space referred to as the "truth." Assigning agent i a constant $\lambda_i \in [0, 1]$ that characterizes the attractiveness of the truth for this agent, serving as a counterpart to the susceptibility in the FJ model, the dynamics are as follows:

$$x_i(t+1) = \lambda_i \frac{1}{|I_i(x(t))|} \sum_{j \in I_i(x(t))} x_j + (1 - \lambda_i)T, \quad (24)$$

where the $I_i(x)$ denote the sets of trusted individuals.

The following result discusses the convergence of the above model

Theorem 2.5. *The opinions of all truth seekers ($\lambda_i < 1$) converge to the truth:*

$$x_i(t) \rightarrow T \quad \text{as } t \rightarrow \infty. \quad (25)$$

The opinions of the remaining agents (with $\lambda_i = 0$) either converge to T or stabilize in finite time at some values $\bar{x}_i$ such that $|\bar{x}_i - T| \geq d$.

The class of "inertial" HK models has been studied, which accounts for the hesitation to change current opinion:

$$x_i(t+1) = (1 - \lambda_i)x_i(t) + \lambda_i \frac{1}{|I_i(x(t))|} \sum_{j \in I_i(x(t))} x_j. \quad (26)$$

Here $\lambda_i \in [0, 1]$ is known as the coefficient of inertia. The agents with $\lambda_i(t) \equiv 0$ are stubborn.

2.1.10 Antagonistic Relations Via Negative Influence

Agreement is not always possible (due to the effects of stubbornness), the agents cooperate in order to reach consensus, always changing their opinions towards each other. In many systems, arising in economics, natural sciences and robotics, such positive (attractive) couplings among the agents coexist with negative (repulsive) couplings, leading to the agents' distancing.

An attempt to persuade a person sometimes shifts his/her opinion away from the persuader's opinion. Abelson refers to this as the "boomerang effect", an unintended consequence of the persuasion.

The theories of balance by Altafini and cognitive dissonance explain the presence of negative influences by coevolution of the opinions and personal relations between individuals. An agreement with negatively evaluated person creates a psychological tension, and individuals can resolve such tensions by "moving their opinions away from a disliked source".

Unlike bounded confidence models, instead of rejecting extremely dissimilar opinions, agents move the opinions away from those having negative influence.

2.1.11 Altafini's model of opinion formation[8]

Altafini's model is a generalization of Abelson's model, accounting for the fact that the matrix $A(t)$ need not be non-negative. The corresponding Laplacian is understood in the modified sense:

$$\dot{x}_i(t) = \sum_{j=1}^n |a_{ij}(t)| (x_j(t) \operatorname{sgn} a_{ij}(t) - x_i(t)) = \sum_{j=1}^n (a_{ij}(t)x_j(t) - |a_{ij}(t)|x_i(t)), \quad \forall i, \quad (27)$$

which is equivalent to:

$$\dot{x}(t) = -L[A(t)]x(t), \quad t \geq 0. \quad (28)$$

Along with the continuous-time model, the discrete-time Altafini model can be considered:

$$x(k+1) = W(k)x(k), \quad x(k) \in \mathbb{R}^n, \quad (29)$$

where the matrix $W| \cdot | = (|w_{ij}(k)|)$ is stochastic and $w_{ii}(k) \geq 0$, while $w_{ij}(k)$ for $i \neq j$ can be negative.

2.1.12 DeGroot-Friedkin Model For Social Power[20]

The concept of social power was first introduced in order to identify the most influential agent in the social network.

The DeGroot-Friedkin (DF) model integrates the opinion formation process with social power evolution through a reflected self-appraisal mechanism. A generalized DF model was proposed and an extension of the DF model to stubborn agents was proposed.

Here, self-appraisal refers to an individual's assessment of one's own capabilities and social power within a network, dynamically evolving with time due to interactions and influence within the network.

Consider $n \geq 2$ individuals discussing a sequence of issues $s = 0, 1, 2, \dots$ in an influence network formulated by a row-stochastic, zero-diagonal relative interaction matrix C . Let $y_i(s, k) \in \mathbb{R}$ denote the opinion of individual i on issue s at time k . $\theta_i \in [0, 1]$ denotes individual i 's susceptibility to interpersonal influence, i.e., $1 - \theta_i$ represents its stubbornness to initial opinion.

Assume that during the discussion of issue s , the self-appraisal of individual i , denoted by $x_i(s) \in [0, 1]$, is static, and each individual forms its opinions according to the FJ model:

$$y_i(s, k+1) = \theta_i \sum_{j=1}^n W_{ij}(s) y_j(s, k) + (1 - \theta_i) y_i(s, 0).$$

Assume that $W_{ii}(s) = x_i(s)$, and $W_{ij}(s) = (1 - x_i(s))C_{ij}$.

This is the DeGroot-Friedkin model for social power with stubbornness incorporated in the model.

Finally, some work on opinion dynamics also analyse the periodic behavior of opinions. Also, instead of considering fixed interactions, some models like the "Deffuant-Weibusch"[9] consider randomized interactions under probability distributions. In such models, known as gossip-based models[11], convergence is studied, both "almost-sure" and "in probability". In some models, the authors see asynchronous updating of opinions as compared to synchronous or simultaneous updating of all opinions at each time[11].

2.2 Game-Theoretic Models For Opinion Formation

The works that use a game theoretic framework for opinion dynamics are highlighted in this section. Each subsection will briefly discuss the techniques that a paper used in modeling. In the game-theoretic approach, all agents try to learn best responses by minimizing a cost function.

2.2.1 Cooperative Control and Potential Games[22]

In cooperative problems such as consensus, agents work together to achieve a common objective. In a discrete-time version of the consensus problem, a group of agents/players $P = \{P_1, \dots, P_n\}$ seeks to come to an agreement, or consensus, upon a common scalar value by repeatedly interacting with one another.

By reaching a consensus, the authors mean converging to the agreement space characterized by

$$a_1 = a_2 = \dots = a_n,$$

where a_i is referred to as the state of agent P_i .

A well-studied protocol, which is referred to here as the "consensus algorithm," can be described as follows:

At each time step $t \in \{0, 1, \dots\}$, each player P_i is allowed to interact with a group of other players, who are referred to as the neighbors of player P_i and denoted as $N_i(t)$.

During an interaction, each player P_i is informed of the current (or possibly delayed) state of all its neighbors. Player P_i then updates its state by forming a convex combination of its state, along with the state of all its neighbors. The consensus algorithm takes on the general form:

$$a_i(t+1) = \sum_{P_j \in N_i(t)} \omega_{ij}(t) a_j(t),$$

where $\omega_{ij}(t)$ is the relative weight that player P_i places on the state of player P_j at time t .

This paper first models consensus as a potential game and then extends it to a weakly acyclic game, proposing learning algorithms for each model.

In the game theoretic approach to cooperative control, each player $P_i \in P$ is assigned an action set A_i and a local objective function $U_i : A \rightarrow \mathbb{R}$, where

$$A = \prod_{P_i \in P} A_i$$

is the set of joint actions.

The challenge of control design in the game-theoretic approach lies in designing both the player objective functions and the learning dynamics, so that players collectively accomplish the objective of the global planner. Learning dynamics will be formulated as a repeated game, in which a one-stage game is repeated at each time step $t \in \{0, 1, 2, \dots\}$.

Learning dynamics in general can have finite memory or unbounded memory. In the latter case, the dynamics of the variable evolving can depend on all its previous values, i.e.

$$\dot{x}(t) = f(x(t-1), x(t-2), \dots, x(0)).$$

The authors study the special case of finite memory learning dynamics where $x(t)$ depends only on the state at $t-1$.

Definition 3. (Potential Games) Player action sets $\{A_i\}_{i=1}^n$, together with player objective functions $\{U_i : A \rightarrow \mathbb{R}\}_{i=1}^n$, constitute a potential game if, for some potential function $\phi : A \rightarrow \mathbb{R}$,

$$U_i(a''_i, a_{-i}) - U_i(a'_i, a_{-i}) = \phi(a''_i, a_{-i}) - \phi(a'_i, a_{-i}) \quad (2)$$

for every player $P_i \in P$, for every $a'_i, a''_i \in A_i$, and for every

$$a_{-i} \in \times_{j \neq i} A_j.$$

This imposes that a player's objective function be appropriately "aligned" with the objective of the global planner.

Definition 4. (Nash Equilibrium) An action profile $a^* \in A$ is called a pure Nash equilibrium if, for all players $P_i \in P$:

$$U_i(a^*_i, a^*_{-i}) = \max_{a_i \in A_i} U_i(a_i, a^*_{-i}). \quad (30)$$

Consensus modelled as a potential game

One can model the consensus problem as one where each player has the objective function:

$$U_i(a_i, a_{-i}) = - \sum_{P_j \in N_i} \|a_i - a_j\|. \quad (31)$$

Here the potential function is:

$$\phi(a) := - \sum_{P_i \in P} \sum_{P_j \in N_i} \frac{\|a_i - a_j\|}{2}$$

In the case where the "interaction graph" induced by neighbor sets $\{N_i\}_{i=1}^n$ is connected, the aforementioned potential function achieves the value of 0 if and only if action profile $a \in A$ constitutes a consensus, i.e.,

$$\phi(a) = 0 \iff a_1 = \dots = a_n.$$

Probabilistic Learning Algorithm For Convergence To Pure Nash Equilibria

The following learning algorithm is called spatial adaptive play(SAP) for consensus modeled as a potential game:

At each time $t > 0$, one player $P_i \in P$ is randomly chosen (with equal probability for each player) and allowed to update its action. All other players must repeat their actions, i.e., $a_{-i}(t) = a_{-i}(t-1)$. At time t , the updating player P_i randomly selects an action from A_i according to its strategy $p_i(t) \in \Delta(A_i)$, where the a_i -th component $p_i^a(t)$ of its strategy is given as

$$p_i^a(t) = \frac{\exp(\beta U_i(a_i, a_{-i}(t-1)))}{\sum_{\bar{a}_i \in A_i} \exp(\beta U_i(\bar{a}_i, a_{-i}(t-1)))},$$

for some exploration parameter $\beta \geq 0$.

In a repeated potential game in which all players adhere to SAP, the stationary distribution $\mu \in \Delta(A)$ of the joint action profiles is:

$$\mu(a) = \frac{\exp(\beta \phi(a))}{\sum_{\bar{a} \in A} \exp(\beta \phi(\bar{a}))}.$$

In the potential game formulation of the consensus problem, the joint actions that maximize the potential function are precisely the consensus points, provided that the interaction graph is connected. Therefore, if all players update their actions using the learning algorithm SAP with sufficiently large β , then the players will asymptotically reach a consensus with arbitrarily high probability.

The above algorithm can be modified for restricted action spaces too, where the feasible action set is a proper subset of the set of all actions.

Weakly acyclic games

Consider any finite game G with player set P , action set A , and utility functions $\{U_i\}_{i=1}^n$. A better reply path is a sequence of action profiles $a_1, a_2, \dots, a_L$ such that, for every $1 \leq \ell \leq L-1$, there is exactly one player P_{i_ℓ} such that

$$a_{i_\ell}^\ell = a_{i_\ell}^{\ell+1}, \quad a_{-i_\ell}^\ell = a_{-i_\ell}^{\ell+1},$$

and

$$U_{i_\ell}(a^\ell) < U_{i_\ell}(a^{\ell+1}).$$

In other words, one player moves at a time, and each time that a player moves, it increases its own utility.

Definition 5. *Game G is weakly acyclic if, for any $a \in A$, there exists a better reply path starting at a and ending at some pure Nash equilibrium of G .*

The authors note that potential games are a special case of weakly acyclic games.

The following proposition links weakly acyclic games to a formulation using potential functions:

Proposition 2.6. *A game is weakly acyclic if and only if there exists a potential function $\phi : A \rightarrow \mathbb{R}$ such that, for any action $a \in A$ that is not a Nash equilibrium, there exists a player $P_i \in P$ with an action $a'_i \in A_i$ such that*

$$U_i(a'_i, a_{-i}) > U_i(a_i, a_{-i}) \quad \text{and} \quad \phi(a'_i, a_{-i}) > \phi(a_i, a_{-i}).$$

An advantage in modeling cooperative control problems as weakly acyclic games as compared to potential games is the flexibility of defining the objective functions. In potential games, one must make sure the objective function aligns precisely with the potential function.

Sometimes Weakly Acyclic games The authors now consider the case when the objective function is time-varying. The authors introduce a state variable $x(t)$ to account for the dynamics of the objective function. Such a state-dependent objective function based formulation is known as a Markov or stochastic game.

In this framework, each player P_i is now assigned a local state-dependent objective function $U_i : A \times X \rightarrow \mathbb{R}$, where X is the set of states. In the consensus problem, X could represent the set of possible (time-varying) interaction topologies. Denote the objective function of player P_i at time t as $U_i(a(t), x(t))$, where $a(t)$ and $x(t)$ are the action profile and state at time t . The state dynamics take on the general form

$$x(t) = f(x(t-1), a(t-1), N(t))$$

where $N(t)$ is nature's influence at time t . For example, one can think of $N(t)$ as time-varying neighborhood sets in the consensus problem.

Definition 6. *An action profile a^* is an invariant Nash equilibrium if*

$$U_i(a^*, x) = \max_{a_i \in A_i} U_i((a_i, a_{-i}^*), x) \quad \forall x \in X.$$

Definition 7. *A game is sometimes weakly acyclic if there exists a potential function $\phi : A \rightarrow \mathbb{R}$ and a finite time constant T such that the following property holds: For any time $t_0 > 0$, if $a(t_0) = a^0$ is not an invariant Nash equilibrium, then there exists a player $P_i \in P$, an action $a'_i \in A_i$, and a time $t \in [t_0, t_0 + T]$, where*

$$U_i((a'_i, a_{-i}^0), x(t)) > U_i(a^0, x(t))$$

and

$$\phi(a'_i, a_{-i}^0) > \phi(a^0),$$

provided that $a(t_0) = a(t_0 + 1) = \dots = a(t - 1)$.

The authors note that a sometimes weakly acyclic game has at least one invariant Nash equilibrium, i.e., any action profile that maximizes the potential function ϕ .

Learning Dynamics for Sometimes Weakly Acyclic Games

The authors define a player's *better reply set* for any action profile $a \in A$ and state $x \in X$ as

$$B_i(a, x) := \{a'_i \in A_i : U_i((a'_i, a_{-i}), x) > U_i(a, x)\}.$$

The *better reply with inertia* dynamics can be described as follows: At each time $t > 0$, each player P_i presumes that all other players will continue to play their previous actions $a_{-i}(t - 1)$. Under this presumption, each player $P_i \in P$ selects an action according to the following strategy at time t :

$$B_i(a(t - 1), x(t)) = \emptyset \Rightarrow a_i(t) = a_i(t - 1)$$

$$B_i(a(t - 1), x(t)) \neq \emptyset \Rightarrow \begin{cases} \Pr[a_i(t) = a_i(t - 1)] = \alpha(t), \\ \Pr[a_i(t) = a'_i] = \frac{(1 - \alpha(t))}{|B_i(a(t - 1), x(t))|} \end{cases}$$

for any action $a'_i \in B_i(a(t - 1), x(t))$, where $\alpha(t) \in (0, 1)$ is the player's inertia at time t . According to these rules, player P_i will stay with the previous action $a_i(t - 1)$ with probability $\alpha(t)$, even when there is a perceived opportunity for improvement.

Assumption 1. *There exist constants $\underline{\epsilon}$ and $\bar{\epsilon}$ such that, for all times $t \geq 0$ and for all players $P_i \in P$,*

$$0 < \underline{\epsilon} < \alpha_i(t) < \bar{\epsilon} < 1.$$

Theorem 2.7. *Consider any n -player sometimes weakly acyclic game with finite action sets. If all players adhere to the better reply with inertia dynamics satisfying the above assumption, then the joint action profiles will almost surely converge to an invariant Nash equilibrium.*

The next two sections discuss papers where the FJ model is interpreted as a best response dynamics where each agent tries to minimise a cost function.

2.2.2 Opinion dynamics in social networks with stubborn agents: Equilibrium and convergence rate[23]

Agents in a social network iteratively update their opinions based on their own initial opinions and observing the opinions of their neighbors, and this process can be viewed as a myopic cost-minimization response (i.e., the so-called best response) to the others' actions. In this paper a non-Bayesian approach is taken. A Bayesian approach to social learning is one in which the opinions evolve based on an underlying like seen in the above paper with Markov games.

The authors of this paper study the convergence of opinions to equilibria, bounds on the convergence rates, and their dependence on network structure and location and level of stubborn agents. They carry out demonstrations on Erdos-Renyi random graphs and small-world graphs.

When there are no stubborn agents, the paper's model reduces to a "continuous coordination game" where the (noisy) best-response dynamics converge to consensus, a common opinion in which the impact of each agent is directly proportional to its degree in the social network.

In the presence of stubborn agents, the opinions don't converge to a consensus, as will be shown. Partially stubborn agents are biased towards their initial opinions and fully stubborn agents don't change their opinion at all.

The paper characterizes the equilibrium in the presence of stubborn agents using hitting probabilities of a random walk over the interaction graph(social network).

Model

Consider a social network with n agents, denoted by a graph $G(V, E)$ where agents are the vertices and edges indicate the pairs of agents that have interactions. For each agent i , define its neighborhood ∂i as the set of agents that node i interacts with, i.e.,

$$\partial i := \{j : (i, j) \in E\}.$$

Each agent i has an initial opinion $x_i(0) \in [0, 1]$. Let

$$x(0) := [x_1(0) \cdots x_n(0)]^T$$

denote the vector of initial opinions. The authors assume each agent i has a cost function of the form

$$J_i(x_i, x_{\partial i}) = \frac{1}{2} \sum_{j \in \partial i} (x_i - x_j)^2 + \frac{1}{2} K_i (x_i - x_i(0))^2,$$

that he tries to minimize, where $K_i \geq 0$ measures the stubbornness of agent i regarding his initial opinion. When none of the agents are stubborn (i.e., all K_i 's are zero), the above formulation defines a *coordination game* with continuous payoffs because any vector of opinions

$$x = [x_1 \cdots x_n]^T$$

with $x_1 = x_2 = \cdots = x_n$ is a Nash equilibrium.

Here, the authors consider a synchronous version of the game between the agents. At each time, every agent observes the opinions of his neighbors and updates his opinion based on these observations and also his own initial opinion in order to minimize his cost function.

It is easy to see, for every agent i , the best-response strategy is

$$x_i(t+1) = \frac{1}{d_i + K_i} \sum_{j \in \partial i} x_j(t) + \frac{K_i}{d_i + K_i} x_i(0),$$

where $d_i = |\partial i|$ is the degree of node i in graph G .

Define a matrix $A \in \mathbb{R}^{n \times n}$ such that

$$A_{ij} = \begin{cases} \frac{1}{d_i + K_i}, & \text{if } (i, j) \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Define a diagonal matrix $B \in \mathbb{R}^{n \times n}$ with

$$B_{ii} = \frac{K_i}{d_i + K_i}, \quad \text{for } 1 \leq i \leq n.$$

Thus, in matrix form, the best-response dynamics are given by

$$x(t+1) = Ax(t) + Bx(0).$$

Iterating the above equation shows that the vector of opinions at each time $t \geq 0$ is

$$x(t) = A^t x(0) + \sum_{s=0}^{t-1} A^s B x(0).$$

Equilibrium Analysis

The paper gives the following results on the equilibrium attained after convergence of the model:

Lemma 2.8. *The best-response dynamics converge to a unique equilibrium where the opinion of each agent is a convex combination of the initial opinions of the stubborn agents.*

The authors construct a modified graph, where the edges of the original graph have a weight of 1 and additional fictitious vertices attached to partially stubborn agents with the edge weight $= K_i$.

The paper shows that the coefficient in the convex combination for agent i corresponding to stubborn agent j are the hitting probabilities of the random walk on the modified graph that starts from i and ends in j , if j is fully stubborn, or u_j which is the fictitious vertex attached to j , which is partially stubborn. Here for the random walk, the transition probabilities are the normalised edge weights.

They also provide an interesting lemma connecting the equilibrium state with electrical networks:

Lemma 2.9. *Consider G as an electrical network where the conductance of each edge is 1 and each stubborn agent i is a voltage source of $x_i(0)$ volts with an internal conductance K_i . Fully stubborn agents are ideal voltage sources with infinite internal conductance (zero internal resistance).*

Then, under the best-response dynamics, the opinion of each agent at equilibrium is just its voltage in the electrical network.

Bounds on convergence times

Define for a non-fully stubborn agent i , $e_i(t) = x_i(t) - x_i(\infty)$ and $\tilde{\pi}_i = \frac{w_i}{Z}$ where Z is a normalizing constant such that $\tilde{\pi}_i$ adds up to 1.

Lemma 2.10. *Let $\|\cdot\|_{\tilde{\pi}}$ be the weighted norm with weights given $\tilde{\pi}$.*

$$\|\tilde{e}(t)\|_{\tilde{\pi}} \leq (\lambda_A)^t \|\tilde{e}(0)\|_{\tilde{\pi}},$$

where λ_A is the largest eigenvalue of A .

The convergence time is defined as

$$\tau(\nu) := \inf\{t \geq 0 : \|\tilde{e}(t)\|_{\tilde{\pi}} \leq \nu\}$$

for some small fixed $\nu > 0$. Therefore,

$$\left(\frac{1}{1 - \lambda_A} - 1\right) \leq \tau(\nu) \log\left(\frac{\|\tilde{e}(0)\|_{\tilde{\pi}}}{\nu}\right) \leq \frac{1}{1 - \lambda_A},$$

so

$$\tau(\nu) = \Theta\left(\frac{1}{1 - \lambda_A}\right)$$

as n grows.

Thus this gives a bound on the convergence time. The authors have analyzed the eigenvalue for different graphs.

2.2.3 How Bad Is Forming Your Own Opinion[24]

In this FOCS 2011 paper by Jon Kleinberg et al, the authors study the social cost of disagreement when consensus is not reached, due to internal preferences of agents which as the authors refer to as "stubbornness". They study the "price of anarchy" of the equilibrium attained on repeated averaging in the Friedkin-Johnson model. The price of anarchy is a measure of how close the outcome of the equilibrium to the "socially optimal" outcome, one that minimizes the social cost.

FJ model of averaging

In a social network represented by a weighted graph $G(V, E)$, each node i maintains a persistent internal opinion s_i which remains constant. If $w_{i,j} \geq 0$ denotes the weight on the edge (i, j) in G , then in one time step, node i updates her opinion to be the average

$$z_i = \frac{s_i + \sum_{j \in N(i)} w_{i,j} z_j}{1 + \sum_{j \in N(i)} w_{i,j}}, \quad (32)$$

where $N(i)$ denotes the set of neighbors of i in G .

The authors can equivalently view nodes' actions in each time step as myopically optimizing a quadratic cost function: updating z_i as in the above equation is the same as choosing z_i to minimize

$$(z_i - s_i)^2 + \sum_{j \in N(i)} w_{i,j} (z_i - z_j)^2. \quad (2)$$

The authors therefore take this as the cost that i incurs by choosing a given value of z_i , so that averaging becomes a form of cost minimization.

The social optimum is the vector of node opinions $y = (y_i : i \in V)$ that minimizes the social cost, defined as the sum of all players' costs:

$$c(y) = \sum_i \left((y_i - s_i)^2 + \sum_{j \in N(i)} w_{i,j} (y_i - y_j)^2 \right).$$

Results

The *price of anarchy* is defined as the ratio between the social cost of the Nash equilibrium and the social optimum.

The authors show that in any undirected graph, the price of anarchy is bounded above by $\frac{9}{8}$. This is partly due to the symmetry of influence weights w_{ij} in the graph.

However, in the directed case, the authors can have unbounded prices of anarchy. Consider an outward star network where the center node, with initial opinion 1 influences all the outer vertices, with initial opinion 0. The optimal social cost is 1, but the cost at equilibrium would be $\Omega(n)$. Hence the price of anarchy is also $\Omega(n)$.

The authors show that a price of anarchy lower-bounded by a polynomial in n can in fact be achieved in directed graphs of constant degree, so this behavior is not simply a consequence of large in-degree.

They also show that for Eulerian asymmetric directed graphs (Eulerian graphs with no oppositely oriented edges (i,j) and (j,i) present) with maximum degree d and edge expansion at most α , a bound of $O(d^2 \alpha^{-2})$ on the price of anarchy. The edge expansion, is defined as the minimum ratio of the number of edges crossing a cut to the size of the smaller partition, taken over all possible partitions of the graph's vertices. Additionally, for unweighted d -regular Eulerian graphs, they show a bound of $+1$ on the price of anarchy. A d -regular directed graph is one where each vertex has in-degree and out-degree both d .

Finally, the authors discuss the problem of modifying the original graph to improve the social cost. They show that in multiple variants, this problem is NP-hard. They develop a $\frac{9}{4}$ -approximation algorithm where one can add edges between any pair of vertices and give an algorithm to find the optimal amount of edge weight to be added to an edge.

2.2.4 Game-Theoretic Analysis Of The Hegselmann Krause model[27]

In this paper by Etasami et al, the authors view the HK dynamics as the best response dynamics to a "network formation game". Additionally, they analyse the convergence of HK dynamics, when updates are made synchronously and asynchronously.

In general the update rule is

$$x(t+1) = A(t, x(t), \varepsilon) x(t)$$

where $A(t, x(t), \varepsilon)$ is an $n \times n$ row-stochastic matrix.

Synchronous and Asynchronous HK models

In the synchronous Hegselmann-Krause model, each agent i updates its value at time $t = 0, 1, 2, \dots$, by averaging its own value and the values of all the other agents that are in its ε -neighborhood at time t .

To be more specific, given a profile $x(t)$ at time t , define the matrix $A(t)$ by

$$A_{ij}(t) = \begin{cases} \frac{1}{|N_i(t)|} & \text{if } j \in N_i(t), \\ 0 & \text{otherwise,} \end{cases}$$

where $N_i(t)$ is the set of agents in the ε -neighborhood of agent i , i.e.,

$$N_i(t) = \{j \in [n] \mid \|x_i(t) - x_j(t)\| \leq \varepsilon\}.$$

The authors give an improved upper bound on the convergence of the synchronous model:

Theorem 2.11. *The termination time of the synchronous Hegselmann-Krause dynamics in arbitrary finite dimensions is independent of the dimension and is bounded from above by*

$$T_n \leq n^8 + n.$$

In the asynchronous case, only one agent, namely i^* , updates its value to the average of its neighbors, while the others remain unchanged. Selection of such an agent may be at random or based on some predefined order. The authors assume that the agents are chosen uniformly at random to update their opinions. In that case, the updating matrix can be written as

$$A_{ij}(t) = \begin{cases} \frac{1}{|N_{i^*}(t)|} & \text{if } i = i^*, j \in N_{i^*}(t), \\ 1 & \text{if } i = j \neq i^*, \\ 0 & \text{otherwise,} \end{cases}$$

where here the authors have assumed that agent i^* updates its opinion at time t .

In the heterogeneous Hegselmann-Krause model, each agent i is able to observe only its ε_i -neighborhood, and the authors have

$$N_i(t, \varepsilon_i) = \{j \in [n] \mid \|x_i(t) - x_j(t)\| \leq \varepsilon_i\}.$$

Network Formation Game

Consider a scenario with n road constructors (players) in $\mathbb{R}^d$, each funded equally by the government. Each player receives a fixed budget of $(n-1)\varepsilon^2$, corresponding to a potential subsidy of ε^2 for each of the $\binom{n}{2}$ possible roads they can construct. The government's goal is to have all possible roads built. To incentivize this, a player is penalized by ε^2 for every road they choose not to construct—meaning the government reclaims the unused subsidy.

However, each player is restricted to constructing roads only within their ε^2 -neighborhood, reflecting a risk-averse nature: they won't spend beyond what is subsidized. Players act myopically, seeking to build roads that offer them the greatest benefit. If two players located at positions $x, y \in \mathbb{R}^d$ collaborate to construct a road, the cost is proportional to the squared Euclidean distance $\|x - y\|^2$. Thus, the utility function for player $i \in [n]$, located at x_i , is given by

$$U_i(x_i, x_{-i}) = (n-1)\varepsilon^2 - \sum_{j=1}^n \min \{\|x_i - x_j\|^2, \varepsilon^2\}, \quad (33)$$

where x_{-i} denotes the actions (locations) of all players except the i -th player.

The following results link the HK dynamics with the game defined above:

Lemma 2.12. *The sequence of the players' best responses in the network formation game under some specific updating scheme is equivalent to the evolution of the asynchronous Hegselmann-Krause dynamics under the same updating scheme.*

Proposition 2.13. *An action profile $(x_1^*, x_2^*, \dots, x_n^*)$ is a Nash equilibrium of the network formation game if and only if it is a steady state of the asynchronous Hegselmann-Krause dynamics.*

2.2.5 Tuning Cooperative Behavior in Games With Nonlinear Opinion Dynamics by Shinkyu Park et al[30]

Sociologists, political scientists, and economists have long argued that reciprocity is key to promoting cooperation. Computer simulations have shown that reciprocal strategies can elicit mutual cooperation in repeated games: the winning strategy for the repeated prisoner's dilemma was Tit-for-Tat (TFT), where an agent reciprocates the opponent's strategy in the previous round; more generally, successful strategies were nice, forgiving, provokable, and clear.

The authors have created a new framework for multi-agent repeated games by using non-linear opinion dynamics. Each person's strategic decisions' evolution depends on payoffs based on models of rationality as well as the agents' observation of the strategies that others play. They also show the "bistability of equilibria" in repeated two-strategy games in which multiple agents observe the opinions of others over a fixed network.

Model

Consider an N_a -agent decision-making problem where each agent selects a strategy, continuously in time t , from the set $\{1, \dots, N_s\}$ of N_s available strategies. Each agent performs a probabilistic choice of strategy where $x_i(t) \in X_i$ is the probability distribution for the strategy selection at time t of agent i , and X_i is the probability simplex in $\mathbb{R}^{N_s}$. The j -th element x_{ij} of x_i is the probability that agent i selects strategy j . Following convention in game theory, x_i is the *mixed strategy* of agent i , and $x = (x_1, \dots, x_{N_a}) \in X$ is the *mixed strategy profile*, where $X = X_1 \times \dots \times X_{N_a}$.

The mixed strategy $x_i(t)$ is defined by the *logit choice function* and depends on agent i 's opinion state at time t , $\bar{z}_i(t) = (\bar{z}_{i1}, \dots, \bar{z}_{iN_s})(t) \in \mathbb{R}^{N_s}$, as follows:

$$x_{ij} = \sigma_j(\bar{z}_i) = \frac{\exp(\eta^{-1} \bar{z}_{ij})}{\sum_{l=1}^{N_s} \exp(\eta^{-1} \bar{z}_{il})}, \quad (34)$$

where the positive constant η is called the *noise level* or *rationality parameter*. Each entry $\bar{z}_{ij}$ of $\bar{z}_i$ represents agent i 's preference for the j -th available strategy. The *relative opinion state* z_i , with j -th entry

$$z_{ij} = \bar{z}_{ij} - \frac{1}{N_s} \sum_{l=1}^{N_s} \bar{z}_{il},$$

defines an agent's preferred strategies.

The authors define the rate-of-change $\dot{\bar{z}}_i = \frac{d\bar{z}_i}{dt}$ of agent i 's opinion state $\bar{z}_i$ in response to payoffs and social interactions, with the continuous-time nonlinear opinion dynamics model:

$$\dot{\bar{z}}_{ij} = -d_i \left(\bar{z}_{ij} - u_i \sum_{k=1}^{N_a} 2R(A_{ik}^j z_{kj}) - U_{ij}(x) \right), \quad (35)$$

with $\bar{z}_i(0) \in \mathbb{R}^{N_s}$. The term $A_{ik}^j \in \mathbb{R}$ is the weight agent i places in its evaluation of strategy j on its observation of agent k 's opinion of strategy j . The constant resistance parameter $d_i > 0$ reflects the speed with which agent i 's opinions change. The attention parameter $u_i > 0$ reflects the weight placed on incentives derived from social interactions, where $R : \mathbb{R} \rightarrow [0, 1]$ is a saturating function.

They define R as the saturating function:

$$R(A_{ik}^j z_{kj}) = \mathbb{P}(A_{ik}^j z_{kj} \geq \varepsilon), \quad (36)$$

where ε is a random variable with a symmetric and unimodal probability density function, e.g., the standard normal distribution.

Results

The authors show that the social interaction R incentivizes each agent to reciprocate with other agents in the strategy selection, and the level of reciprocation is determined by the attention parameter u and the number n_j^* of agents preferring the same strategy under consideration.

Under this model, the authors show that each agent i decides to cooperate (or defect) if its opponent does so in the previous stage. This behavior resembles TFT, a well-known reciprocity-based strategy in discrete-time iterated games.

2.2.6 Open Loop equilibrium strategies for dynamic influence maximization game by Etasami et al[26]

In advertising campaigns, the evolution of public opinion comprises of two factors: i) the advertising or investment strategies of external influencers, such as companies or political leaders (referred to as players); and ii) the internal interactions among social entities (referred to as individuals). A major challenge for the players is to deploy investment strategies to shift public opinion toward their objectives, subject to certain budget constraints.

Opinion influence maximization has been extensively studied in the past literature, mainly using static optimization or static games, but not in dynamic or evolutionary games.

The authors consider a dynamic hybrid game and under certain convexity assumptions, obtain an efficient algorithm for solving its "open-loop equilibrium" strategies.

Model

Consider a set $[n] = \{1, \dots, n\}$ of individuals that interact over a directed social network $G = ([n], E)$, and a set $[m] = \{1, \dots, m\}$ of influencers (players). The authors use $\beta_j \geq 0$ to denote the advertising budget of player $j \in [m]$. At any time $t \geq 0$, the authors use an m -dimensional row vector $\mathbf{x}_i(t) := (x_{i1}(t), \dots, x_{im}(t)) \in [0, 1]^m$ to denote individual i 's opinion on the players, where $x_{ij}(t)$ is the opinion of individual i about player j at time t . The authors also use the matrix $\mathbf{x}(t) \in [0, 1]^{n \times m}$ to refer to the opinions of all the individuals at time t .

The authors assume that there are K advertising campaign times $0 \leq t_1 < \dots < t_K$, at which the individuals undergo the influence of the players. Moreover, the authors define $t_0 = 0$ and t_{K+1} to be the initial time and the termination time of the game. Let us use $\beta_j(t_k)$ to denote the remaining advertising budget of player j at campaign time t_k , where the authors note that $\beta_j(t_1) = \beta_j$. At campaign time t_k , $k \in [K]$, each player j invests according to the (action) vector $\mathbf{b}_j(k) = (b_{1j}(k), \dots, b_{nj}(k))'$, where $b_{ij}(k)$ is the amount that player j invests on individual i at campaign time t_k .

In particular, the action space for player j at time t_k is defined as

$$B_j(k) := \left\{ \mathbf{b} \in \mathbb{R}_+^n : \sum_{i=1}^n b_i \leq \beta_j(k) \right\}.$$

In the absence of the players, i.e., during time intervals $\bigcup_{k=1}^{K+1} (t_{k-1}, t_k)$, the authors assume that individuals' opinions evolve according to the Abelson model, which is given by the ordinary differential equation

$$\dot{\mathbf{x}}(t) = -L\mathbf{x}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0.$$

Here, $\mathbf{x}_0$ is the individuals' initial opinions, and L is the weighted Laplacian matrix corresponding to the social network $G = ([n], E)$.

As a result of the advertising campaigns at time instances $\{t_k\}_{k=1}^K$, the opinions of the individuals undergo a state jump, which obeys the hybrid model:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= -L\mathbf{x}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0, \quad \forall t \in [t_0, t_{K+1}] \setminus \{t_k\}_{k=1}^K \\ \mathbf{x}_{ij}(t_k^+) &= \varphi(x_{ij}(t_k), b_{i1}(k), \dots, b_{im}(k)) \quad \forall i, j, k \in [K]. \end{aligned}$$

Here, t_k^+ is an infinitesimal time after t_k , and the jump function $\varphi : \mathbb{R}_+^{m+1} \rightarrow [0, 1]$ captures the change in individuals' opinions as a result of players' investment strategies at campaign times.

The authors use 2 different jump functions,

- The first case is for a single player, in which the authors can drop the dependency on j and write

$$\varphi(x_i(t_k), b_i(k)) = x_i(t_k) + b_i(k).$$

where $b_i(k) \in [0, 1 - x_i(t_k)]$ to assure that the opinions always remain in the interval $[0, 1]$

- In a multiplayer game, the impact of any player's budget allocation on some individual i will be reduced as other players allocate budget to i . Hence, the authors assume that the impact of budget allocation on the individuals' opinions is normalized and given by

$$\phi(x_{ij}(t_k), b_{i1}(k), \dots, b_{im}(k)) = x_{ij}(t_k) + \frac{b_{ij}(k)}{1 + \sum_{\ell=1}^m b_{i\ell}(k)}.$$

Each player j aims to maximize its average payoff expressed in terms of individuals' opinions:

$$U_j(b_j, b_{-j}) = \frac{1}{K+1} \sum_{k=1}^{K+1} u_j(x_j(t_k), b_j(k), k)$$

where $x_j(t_k) = (x_{1j}(t_k), \dots, x_{nj}(t_k))'$ is the vector of individuals' opinions about player j at time t_k , and $b_j := (b_j(1), \dots, b_j(K))$ is the investment strategy of player j (by convention, $b_j(K+1) = 0$). The functions $u_j(\cdot, k) : \mathbb{R}_+^{2n} \rightarrow \mathbb{R}_+$, for $k \in [K+1]$, are the stage utilities for player j . The goal of each player j is to choose an investment strategy b_j to maximize its average payoff subject to its budget constraint.

Definition 8. An investment strategy profile $(b_1, \dots, b_m)$ is called an open-loop equilibrium if each player observes the initial opinions and budgets $x(t_0), \beta(t_0)$, and chooses an investment strategy sequence that satisfies the equilibrium conditions, i.e.,

$$U_j(b_j, b_{-j}) \geq U_j(\hat{b}_j, b_{-j})$$

for any player j and any admissible investment strategy $\hat{b}_j$.

The following results show how solving an optimization problem for getting the optimal investment strategy for single and multi player games:

Result For A Single Player Game

To obtain the optimal investment strategy for the player, one needs to solve the following optimization problem:

$$\begin{aligned} \max \quad & \frac{1}{K+1} \sum_{k=1}^{K+1} u(x(t_k), b(k), k) \\ \text{subject to} \quad & \dot{x}(t) = -Lx(t), \quad x(t_0) = x_0, \quad t \in [t_0, t_{K+1}] \setminus \{t_k\}_{k=1}^K, \\ & x(t_k^+) = x(t_k) + b(k), \quad \forall k \in [K], \\ & b(k) \leq \mathbf{1} - x(t_k), \quad \forall k \in [K], \\ & \sum_{k=1}^K \mathbf{1}^\top b(k) \leq \beta, \quad b(k) \in \mathbb{R}_+^n, \quad \forall k \in [K], \end{aligned} \tag{37}$$

where $\mathbf{1}$ is the column vector of all ones. This problem can be solved as a concave program.

2.2.7 Influence of Conformist and Manipulative Behaviors on Public Opinion[31]

This paper by Etasami et al. discusses two common tendencies of people:

- **Conformity:** Tendency of people to express their opinion close to public opinion
- **Manipulation:** Tendency of people to manipulate their publicly expressed opinion so that the public opinion becomes close to their actual opinion

The paper gives some real life scenarios where such behavior is expressed:

- **Recommendation Systems:** Instances of manipulative behavior can be seen in the reviews that users give on different rating platforms. For example some users have deliberately given lower ratings (than the actual rating that they would give) to popular films to average out the emotional (fan-favourite) ratings that others give and bring the average rating of the film closer to their true opinion.
- **Political Elections:** In political elections, many individuals often agree with neither candidate, but they vote somehow to bring public opinion closer to their own actual belief.

Contributions

This paper investigates a dynamical opinion network where each individual possesses a personal belief, modeled as a fixed scalar in the interval $[0, 1]$, concerning a specific subject. At each discrete time step, individuals express opinions which may deviate from their true beliefs. The authors focus on two social behaviors that lead to such deviations: *conformity* and *manipulation*. The primary objective is to analyze the roles these behaviors play in shaping public opinion, defined as the arithmetic mean of all expressed opinions at any given time.

The paper further examines the divergence between public opinion and the average of the actual beliefs, which would represent the public opinion in a fully truthful society. This work extends previous research by analyzing scenarios where expressed opinions are drawn from a continuum rather than a binary or finite set. Additionally, the model accounts for individuals who are *truthful*, consistently expressing their actual beliefs. A generalized game-theoretic framework is developed to model the network of individuals and their social behaviors, incorporating truthfulness as a balance between conformity and manipulation.

The authors introduce and explore various single and multistage games where individuals are modeled as players and their expressed opinions serve as their actions. The cost functions are carefully defined to simultaneously capture conformist and manipulative tendencies. The Nash equilibria of each game are characterized, with analysis of convergence and convergence rates of the players' actions.

In earlier sections, the focus is on binary expressed opinions—either approval or disapproval—mimicking responses in opinion polls such as tracking surveys. First they discuss a game with only manipulative individuals then they incorporate conformity.

Recognizing the limitations of binary models for more expressive social contexts, the later sections generalize the framework to continuous opinion values in $[0, 1]$. The study considers a generalized game over an undirected network, where each individual perceives public opinion as the average of her neighbors' previously expressed opinions. The paper analyzes opinion evolution under both single-stage and multistage games. Notably, it is shown that when individuals minimize their perceived regret in the multistage game, their cost aligns with that of the Nash equilibrium in the single-stage game.

3 Opinion Dynamics for Utility Maximizing Agents: Exploring the Impact of the Resource Penalty- Wankhede et al[1]

3.1 Contributions Of the Paper

This paper defines a social network, a directed graph based model, where each agent has her own internal preferences, and is also influenced by others. Each agent has a utility function which she tries to maximize via the evolving dynamics of her own opinion. She does this by a *gradient ascent* of her utility function. The novelty introduced in this paper is the introduction of a resource penalty term which constrains the opinion from having large magnitudes of opinions. In this setting, the authors analyze the asymptotic behavior and equilibria of the dynamics. Additionally, they consider the special cases with no and "weak" antagonistic or negatively influential relations. They analyze how optimal is a Nash equilibrium, by defining egalitarian and utilitarian prices of anarchy. Finally, they analyze the oscillatory behavior of a 2 person dynamics.

3.2 Modeling And Problem Setup

Consider a set $V := 1, \dots, n$ of n agents, with heterogeneous resources, that form opinions on a single topic, which is about the quantity of resources to use for a certain purpose.

The authors denote the expressed opinion of agent $i \in V$ at time t on the topic as $z_i(t) \in \mathbb{R}$.

The utility function for each agent $i \in V$ is given by:

$$U_i(z, p_i) = -\frac{w_i}{2}(z_i - p_i)^2 - \sum_{k \in V} \frac{a_{ik}}{2}(z_k - z_i)^2 - \frac{1}{4r_i}z_i^4$$

where z is the complete opinion profile, $w_i \in \mathbb{R}_{\geq 0}$ is the internal weight given to self preference, $a_{ij} \in \mathbb{R}$ is the influence of agent j on i , $p_i \in \mathbb{R}$, the internal self preference, which is assumed to be fixed with time and $r_i \in \mathbb{R}_{\geq 0}$, the introduced term which quantifies the amount of resources that i has. The authors assume $a_{ii} = 0$ as there are other terms capturing self preference. The social influence graph $G(V, A)$ is a directed graph with vertices V and adjacency matrix A . If $a_{ij} \neq 0$, there exists a directed link from j to i with weight a_{ij} .

Definition 9 (Neighbor set of an agent). *For each agent $i \in V$, the set*

$$N_i^e := \{k \in V \setminus \{i\} \mid a_{ik} < 0\}$$

denotes the set of its enemies, and the set

$$N_i^f := \{k \in V \setminus \{i\} \mid a_{ik} > 0\}$$

denotes the set of its friends. Further,

$$N_i := N_i^e \cup N_i^f = \{k \in V \setminus \{i\} \mid a_{ik} \neq 0\}$$

denotes the set of neighbors of agent $i \in V$.

The corresponding dynamics obtained from gradient ascent of the utility function are:

$$\dot{z}_i = -w_i(z_i - p_i) + \sum_{k \in V} a_{ik}(z_k - z_i) - \frac{z_i^3}{r_i}$$

The authors can rewrite the above as:

$$\dot{z}_i = f_i(z, p_i) := S_i(z_i, p_i) + C_i(z), \quad \forall i \in V \quad (38)$$

$$S_i(z_i, p_i) = -w_i(z_i - p_i) - \frac{z_i^3}{r_i} \quad (39)$$

$$C_i(z) = \sum_{k \in V} a_{ik}(z_k - z_i) \quad (40)$$

where $S_i(z_i, p_i)$ is the self function and $C_i(z)$, the crowd function that depends on the deviation from its neighbors' opinions.

The function $S_i(\cdot, p_i)$ is continuous and strictly decreasing, with

$$\lim_{z_i \rightarrow -\infty} S_i(z_i, p_i) = \infty \quad \text{and} \quad \lim_{z_i \rightarrow +\infty} S_i(z_i, p_i) = -\infty.$$

Thus, $S_i(\cdot, p_i)$ has exactly one real root for every fixed value of $p_i \in \mathbb{R}$. Let us denote the real root of $S_i(\cdot, p_i)$ as $m_i(p_i) \in \mathbb{R}$, i.e.,

$$S_i(m_i(p_i), p_i) = 0.$$

The authors note that from the values of the self-function $S_i(0, p_i)$ and $S_i(p_i, p_i)$, it follows that

$$0 \leq |m_i| \leq |p_i| \quad \text{and} \quad m_i p_i \geq 0.$$

The authors consider these special cases and in further sections study the dynamics under these assumptions:

Assumption A (No antagonistic relations):

$$\forall i \in V, \quad N_i^e = \emptyset$$

Assumption B (Weak antagonistic relations): For each agent $i \in V$,

$$w_i > \sum_{k \in N_i^e} 2|a_{ik}|,$$

where N_i^e is the set of enemies of agent i .

The authors note that no antagonistic relations is a special case of weak antagonistic relations.

3.3 Asymptotic Behavior Analysis

The authors denote the set of equilibrium points of the dynamics as a function of preferences p as

$$\mathcal{E}(p) := \{z \in \mathbb{R}^n \mid \dot{z} = f(z, p) = 0\}.$$

Lemma 3.1 (Convergence of opinions of socially closed agents). *Let $z(t)$ denote the solution of the dynamics from an initial condition $z(0)$. Then,*

$$\lim_{t \rightarrow \infty} z_i(t) = m_i \quad \forall i \in V_{\text{cl}},$$

where V_{cl} is the set of socially closed agents, i.e.,

$$V_{\text{cl}} := \{i \in V \mid N_i = \emptyset\} \subseteq V.$$

Proof. For a socially closed agent i , the cross term in the dynamics becomes zero due to the absence of neighbors, i.e., $N_i = \emptyset$. Thus, the dynamics of z_i evolve independently.

Define the Lyapunov function:

$$V(z_i) = (z_i - m_i)^2.$$

Taking the time derivative along trajectories of the system gives:

$$\dot{V}(z_i) = 2(z_i - m_i)\dot{z}_i.$$

Substituting the dynamics of z_i and noting that m_i is the unique equilibrium point of $S_i(z_i, p_i) = 0$, one can show:

$$\dot{V}(z_i) < 0 \quad \text{for } z_i \neq m_i.$$

Hence, $V(z_i)$ decreases monotonically, and $z_i \rightarrow m_i$ exponentially as $t \rightarrow \infty$, proving convergence. $\square$

In the case of weak antagonistic relations, the authors get the following result:

Theorem 3.2 (Existence and uniqueness of equilibrium points). *Assuming **weak antagonistic relations**, the dynamics converges to a unique globally exponentially stable equilibrium point $\mathbf{z}^* \in \mathcal{E}$.*

Proposition 3.3. *Let $m_{\min}$ and $m_{\max}$ denote the minimum and maximum of m_i 's. Let $\mathcal{M} = [m_{\min}, m_{\max}]$. Assuming **no antagonistic relations**, the unique equilibrium point $\mathbf{z}_*(\mathbf{t})$ lies in $\mathcal{M}^n$.*

The equilibrium point lies in the interior of $\mathcal{M}^n$ if and only if:

$$\begin{aligned} &\forall i \in V_{\max}, \exists \text{ a directed walk in } G \text{ starting from } j \in V \setminus V_{\max} \text{ to } i, \\ &\text{and } \forall i \in V_{\min}, \exists \text{ a directed walk in } G \text{ starting from } j \in V \setminus V_{\min} \text{ to } i. \end{aligned}$$

Proof Sketch: The first part is a consequence of the fact that otherwise, $\exists i$ such that $\dot{z}_i \neq 0$. Any point outside $\mathcal{M}^n$ cannot be an equilibrium point. The rest follows from the above lemma.

The second part follows from:

$$\forall i \in V, \quad z_i^* = m_{\max} \text{ if and only if } m_i = m_{\max} \text{ and } z_j^* = m_{\max}, \quad \forall j \in N_i.$$

Hence, the equilibrium point lies in the interior if and only if

$$\forall i \in V_{\max}, \exists \text{ a directed walk in } G \text{ starting from } j \in V \setminus V_{\max} \text{ to } i.$$

Similarly, by extending the argument,

$$\forall i \in V_{\min}, \exists \text{ a directed walk in } G \text{ starting from } j \in V \setminus V_{\min} \text{ to } i.$$

In general, the authors might not guarantee convergence. But the authors can get ultimate boundedness of opinions, as a compromise.

Theorem 3.4 (Ultimate boundedness of opinions). *Let $z(t)$ be the solution to the dynamics from the initial condition $z(0)$. Then there exists $\eta \geq 0$ (independent of $z(0)$) and $T(z(0)) \geq 0$ such that*

$$|z_i(t)| \leq \eta, \quad \forall i \in V, \forall t \geq T.$$

Additionally, there exists a set $\Omega \subseteq \mathbb{R}^n$ which is convex, compact, and positively invariant under the dynamics.

3.4 Consensus, Nash Equilibria, and Price of Anarchy

Consensus Equilibria

When everyone has the same equilibrium opinion, then $\mathbf{z}_*(t) = \xi \mathbf{1}$ for some $\xi \in \mathbb{R}$, it's called a consensus equilibrium. The following result is on the existence of a consensus equilibrium.

Theorem 3.5. *(Necessary and sufficient conditions for the existence of a consensus equilibrium.) $\xi_1 \in \mathcal{E}$ if and only if $m_i = \xi$, $\forall i \in V$.*

Proof. The cross term $C_i(\xi \mathbf{1}) = 0$. Hence $\xi_1 \in \mathcal{E}$ if and only if $S_i(\xi) = 0$, $\forall i \in V$, i.e. $m_i = \xi$, $\forall i \in V$. $\square$

Hence the authors can see that if $p_i = 0 \forall i \in V$, then the only possible consensus is the neutral consensus. Even if $p_i = 0$ for some $i \in V$, then the unique root $m_i = 0$ and the only possible consensus is the neutral consensus.

The next result talks about dominance in consensus, which is related to the deviation of one's equilibrium opinion from one's internal preference.

Proposition 3.6. *(Consensus deviation from preference.) Suppose that $p_i \neq 0$, $\forall i \in V$. For each agent $i \in V$, define $\sigma_i = w_i r_i$ and $\Delta_i(\xi) = |p_i - \xi|$. If $\xi \mathbf{1} \in \mathcal{E}$, with $\xi \in \mathbb{R}$, then*

$$\sigma_i \Delta_i(\xi) = \sigma_j \Delta_j(\xi), \quad \forall i, j \in V.$$

In particular, $\sigma_i > \sigma_j$ if and only if $\Delta_i(\xi) < \Delta_j(\xi)$.

Proof. Since $p_i \neq 0$, $\forall i \in V$, the authors also have $0 < |m_i| < |p_i| \forall i \in V$. Then by the above theorem, $\xi \neq 0$. Further from the above theorem, the authors get $r_i S_i(\xi) = 0$, $\forall i \in V$, which implies

$$\sigma_i (p_i - \xi) = \xi^3 = \sigma_j (p_j - \xi), \quad \forall i, j \in V.$$

Since $\sigma_i > 0$, $\forall i \in V$, the result follows. $\square$

The authors call, for all $i \in V$, the scalar σ_i the *consensus dominance weight* of agent i . The consensus dominance weight is directly proportional to the importance weight an agent assigns to its preference and the resources available to it. Thus, an agent with very high resources can exert more influence even if it gives less importance weight to its internal preference. On the other hand, if an agent has lower resources, then it must assign a much higher importance weight to its preference in order to exert more influence in the group.

Nash Equilibria

Every agent is interested in maximizing its utility U_i by suitably choosing its opinion z_i . Thus, this gives rise to a strategic form game $G = \langle V, (\mathbb{R})_{i \in V}, (U_i)_{i \in V} \rangle$ among the set of agents V , with the strategy of agent $i \in V$ being its opinion $z_i \in \mathbb{R}$ and its utility function being $U_i(\cdot)$.

Definition 10. *Let z_{-i} denote the opinions of all agents other than i . Then, the set of Nash equilibria of the game G is given by*

$$\mathcal{NE}(p) := \{z^* \in \mathbb{R}^n \mid \forall i \in V, U_i(z_i^*, z_{-i}^*, p_i) \geq U_i(z_i, z_{-i}^*, p_i), \forall z_i \in \mathbb{R}\}.$$

The Nash equilibrium profile is a "best response" provided other agents play their Nash equilibrium profile. Every agent updates her opinion by the gradient ascent, the partial derivative of their utilities with respect to their opinion. Thus agents revise their opinions towards a local best response, which motivates the next definition:

Definition 11 (Local Nash Equilibrium). A strategy profile $z^* \in \mathbb{R}^n$ is said to be a local Nash equilibrium if and only if $\forall i \in V$, there exists $\rho_i \in \mathbb{R}_{>0}$ such that

$$U_i(z_i^*, z_{-i}^*, p_i) \geq U_i(z_i, z_{-i}^*, p_i), \quad \forall z_i \text{ such that } |z_i^* - z_i| \leq \rho_i.$$

The set of local Nash equilibria of G is denoted by $\mathcal{NE}_\ell(p)$.

It is easy to see that a Nash equilibrium is also a local Nash equilibrium and hence $\mathcal{NE}(p) \subseteq \mathcal{NE}_\ell(p)$.

Lemma 3.7 (Local Nash equilibrium is an equilibrium of the opinion dynamics). If an opinion profile z^* is such that $z^* \in \mathcal{NE}_\ell$, then $z^* \in \mathcal{E}$.

Proof. Since $z^* \in \mathcal{NE}_\ell$, it implies that $\forall i \in V$, z_i^* locally maximizes $U_i(\cdot, z_{-i}^*)$. Thus, the partial derivative of $U_i(\cdot)$ with respect to z_i , evaluated at (z_i^*, z_{-i}^*) , is zero. The claim then follows immediately. $\square$

The following theorem gives a necessary condition and a sufficient condition for an equilibrium of the dynamics to be a local Nash equilibrium.

Theorem 3.8 (Conditions for an equilibrium point of the dynamics to be a local Nash equilibrium). Consider the set of equilibrium points $\mathcal{E}$. Let $z^* = (z_i^*, z_{-i}^*) \in \mathcal{E}$. Then, $z^* \in \mathcal{NE}_\ell$ only if for each $i \in V$,

$$(z_i^*)^2 \geq \tau_i := \frac{r_i}{3} \left[\sum_{k \in \mathcal{N}_i^e} |a_{ik}| - \sum_{k \in \mathcal{N}_i^f} a_{ik} - w_i \right]. \quad (41)$$

Moreover, if the inequality (41) is strict for all $i \in V$, then $z^* \in \mathcal{NE}_\ell$.

Proof. Let the hypothesis be true. By definition, $z^* = (z_i^*, z_{-i}^*)$ is a local Nash equilibrium if and only if $\forall i \in V$, z_i^* locally maximizes $U_i(\cdot, z_{-i}^*)$.

Now since $z^* \in \mathcal{E}$, it is clear that for each $i \in V$, z_i^* satisfies the first-order necessary conditions for it to be a local maximizer of $U_i(\cdot, z_{-i}^*)$.

Now, suppose that $z^* \in \mathcal{NE}_\ell$. Then, the authors have that

$$\left. \frac{\partial^2}{\partial z_i^2} U_i(z_i, z_{-i}^*) \right|_{z_i^*} = \frac{3}{r_i} [\tau_i - (z_i^*)^2] \leq 0, \quad \forall i \in V.$$

This proves the necessary condition in (41). Finally, note that if the inequality is strict for a $z^* \in \mathcal{E}$, then by the second-order sufficiency condition for a point to be a local maximizer, z_i^* is a local maximizer of $U_i(\cdot, z_{-i}^*)$ for each $i \in V$. $\square$

Below is an intuitive result on the non-existence of local Nash equilibria.

Corollary 3.9 (Non-existence of local Nash equilibria). Suppose $\eta_i > 0$ is an ultimate bound on z_i for any $i \in V$. If there exists $i \in V$ such that $\tau_i > \eta_i^2$, with τ_i as defined above, then $\mathcal{NE}_\ell = \emptyset$.

Proof. $\exists T > 0$, $z_i^2(t) \leq \eta_i^2$. Since all local Nash equilibria are equilibria of the dynamics, it must follow that $(z_i^*)^2 \leq \eta_i^2$ where z_i^* is a local Nash equilibrium. The claim follows since the second order condition is violated. $\square$

Theorem 3.10 (Equality of equilibria sets and uniqueness of Nash equilibrium). Suppose for each agent $i \in V$, $\tau_i \leq 0$. Then, $\mathcal{NE} = \mathcal{NE}_\ell = \mathcal{E}$. Moreover if the assumption of weak antagonistic relations holds, then $\mathcal{E}$ is a singleton set.

Proof. Suppose that $\tau_i \leq 0$, $\forall i \in V$. First, it is obvious that $\mathcal{NE} \subseteq \mathcal{NE}_\ell$ and since local Nash equilibria are equilibria of the dynamics, $\mathcal{NE}_\ell \subseteq \mathcal{E}$. Next, the authors show that $\mathcal{E} \subseteq \mathcal{NE}_\ell$. Consider an equilibrium $z^* \in \mathcal{E}$. It can be easily verified that for each $i \in V$, $U_i(\cdot, z_{-i}^*)$ is strictly concave in z_i . Since $z^* \in \mathcal{E}$, the first-order necessary condition for a point to be a local maximizer of $U_i(\cdot, z_{-i}^*)$ is satisfied for each i . Hence, $z^* \in \mathcal{NE}_\ell$.

Finally, the authors show that $\mathcal{NE}_\ell \subseteq \mathcal{NE}$. Since $\forall i \in V$, $U_i(\cdot, z_{-i}^*)$ is a strictly concave function for each z_{-i}^* , the implication follows directly from the definitions of $\mathcal{NE}$ and $\mathcal{NE}_\ell$. This completes the proof of the first part of the claim.

Now suppose that the assumption of weak antagonistic relations holds. Then again $\tau_i < 0$, $\forall i \in V$. Uniqueness of Nash equilibrium now follows from the uniqueness and the first part of the claim. $\square$

Price Of Anarchy

The authors change the objective from utility maximization to cost minimization in this section. Define the cost function for each agent $i \in V$ as

$$\chi_i(z, p) := -U_i(z, p).$$

For the cost function to be non-negative, in this subsection, the authors assume that all inter-agent relations are non-antagonistic.

The authors consider the price of anarchy for two of the most commonly used objective functions in game theory literature, namely, the "egalitarian cost" and the "utilitarian cost", defined as

$$C_E(z, p) := \max_{i \in V} \chi_i(z, p), \quad C_U(z, p) := \sum_{i \in V} \chi_i(z, p).$$

They are common measures in game theory literature to measure the inefficiency of the Nash equilibrium.

For the remaining part of this section, the authors will consider the assumption of **non-zero preferences**, i.e. $\forall i \in V, p_i \neq 0$.

Finally, at the end the authors will look at the case when there are zero preferences.

Definition 12 (Price Of Anarchy). *Consider the opinion formation game G along with the opinion dynamics, and $\mathcal{NE}$, the set of its Nash equilibria. Suppose the "no antagonistic relations" assumption holds and $\mathbf{p} \neq \mathbf{0}$. The "egalitarian price of anarchy" π_E and the "utilitarian price of anarchy" π_U are defined as*

$$\pi_E := \frac{\max_{z \in \mathcal{NE}} C_E(z)}{\min_{z \in \mathbb{R}^n} C_E(z)} \geq 1, \quad \pi_U := \frac{\max_{z \in \mathcal{NE}} C_U(z)}{\min_{z \in \mathbb{R}^n} C_U(z)} \geq 1.$$

π_e compares the cost incurred by the worst performing agent at the worst Nash equilibrium to the minimum possible cost C_E , while π_u compares the total cost incurred by all agents at the worst Nash equilibrium to the minimum possible cost C_U . The closer the value of the prices of anarchy is to unity, the better the quality of the Nash equilibrium. The authors define the *satisfaction ratio* at opinion profile z for every agent $i \in V$ in order to compare the cost incurred by agent i at z and the minimum possible cost it could incur.

Definition 13 (Satisfaction Ratio). *Consider the cost function $\chi_i(z)$ of each agent $i \in V$. Suppose there are no antagonistic relations and all internal preferences are non-zero. The "satisfaction ratio" SR_i for any agent $i \in V$ at any $z \in \mathbb{R}^n$ is defined as*

$$SR_i(z) := \frac{\chi_i(z)}{\min_{z \in \mathbb{R}^n} \chi_i(z)} \geq 1.$$

The following lemma talks about the convexity of the cost function of each agent:

Lemma 3.11 (Convexity of cost function). *Assuming no antagonistic relations, for each $i \in V$, the cost function $\chi_i(\cdot)$ is convex with $m_i \mathbf{1}$ as one of its minimizers.*

Proof. Note that the Hessian of $\chi_i(\mathbf{z})$, $H = [H_{ij}]$ has

$$H_{ii} = (w_i + \frac{3z_i^2}{r_i} + \sum_{k \in [n], k \neq i} a_{ik}),$$

$$H_{jj} = a_{ij}, \forall j \neq i,$$

$$H_{ij} = H_{ji} = -a_{ij}, \forall j \neq i.$$

All other terms are 0. Hence the authors get that by the Gershgorin circle theorem, H is positive semi-definite, meaning the cost function $\chi_i(\mathbf{z})$ is convex $\forall \mathbf{z}$. $m_i \mathbf{1}$ satisfies the first-order necessary condition for a point to be a local minimizer of $\chi_i(\mathbf{z})$. Since $\chi_i(\mathbf{z})$ is convex, $m_i \mathbf{1}$ is also a minimizer of $\chi_i(\mathbf{z})$. □

As seen, in the absence of antagonistic relations, the game has a unique Nash equilibrium $z^* \in \mathcal{NE}$. In the next result, the authors use the satisfaction ratios to give an upper bound on π_e and π_u .

Theorem 3.12 (Bounds on Prices of Anarchy). *Suppose there are no antagonistic relations and all internal preferences are non-zero. Let $z^* \in \mathcal{NE}$ denote the unique Nash equilibrium of the game underlying the opinion dynamics. Then,*

$$1 \leq \pi_E \leq \max_{i \in V} SR_i(z^*), \quad 1 \leq \pi_U \leq \sum_{i \in V} SR_i(z^*).$$

Proof. The authors get the lower bound on π_e and π_u from their definitions. To get an upper bound on π_e , note that

$$\min_{z \in \mathbb{R}^n} \max_{i \in V} \chi_i(z) \geq \max_{i \in V} \min_{z \in \mathbb{R}^n} \chi_i(z).$$

Using this inequality and convexity of $\chi_i(\mathbf{z})$, the authors can upper bound π_e as, $\forall j \in V$,

$$\pi_e \leq \frac{\max_{i \in V} \chi_i(z^*)}{\max_{i \in V} \chi_i(m_i \mathbf{1})} \leq \max_{i \in V} \left(\frac{\chi_i(z^*)}{\chi_j(m_j \mathbf{1})} \right),$$

for any $j \in V$. Choosing $j = i$, from the definition of satisfaction ratios and convexity of $\chi_i(\mathbf{z})$ implies the bound on π_e .

From convexity of $\chi_i(\mathbf{z})$, the authors have

$$\min_{z \in \mathbb{R}^n} \sum_{i \in V} \chi_i(z) \geq \sum_{i \in V} \chi_i(m_i \mathbf{1}).$$

Using this the authors get, $\forall j \in V$,

$$\pi_u \leq \frac{\sum_{i \in V} \chi_i(z^*)}{\sum_{i \in V} \chi_i(m_i \mathbf{1})} = \frac{\sum_{i \in V} \chi_i(z^*)}{\sum_{j \in V} \chi_j(m_j \mathbf{1})} \leq \sum_{i \in V} \left(\frac{\chi_i(z^*)}{\chi_j(m_j \mathbf{1})} \right).$$

Choosing $j = i$, the authors get the bound on π_u needed. □

Corollary 3.13 (PoA is unity for a non-neutral consensus equilibrium). *Suppose there are no antagonistic relations and all internal preferences are non-zero. Suppose the unique $z^* \in \mathcal{NE} = \mathcal{E}$ is a non-neutral consensus equilibrium, i.e., $z^* = m \mathbf{1}$ for some $m \neq 0$. Then, $\pi_e = \pi_u = 1$.*

Proof. From the convexity of cost function and definition of satisfaction ratio, it follows that, $SR_i(m \mathbf{1}) = 1$, $\forall i \in V$. From the bounds on PoA, the authors thus have $\pi_e = 1$. From the proof of the bound on π_u , the authors have

$$1 \leq \pi_u \leq \frac{\sum_{i \in V} \chi_i(m \mathbf{1})}{\sum_{i \in V} \chi_i(m \mathbf{1})} = 1,$$

since $z^* = m \mathbf{1}$ and $m_i = m$, $\forall i \in V$ from the necessary conditions for a consensus equilibrium. □

It concludes that PoA is unity for a non-neutral ($p_i \neq 0$) consensus equilibrium.

A neutral consensus is socially optimal. Assume no antagonistic relations and $\mathbf{p} = 0$. As shown, the unique equilibrium point $\mathbf{z}^* = 0 \in E = NE$. From the convexity of $\chi_i(\mathbf{z})$, 0 is a minimizer of $\chi_i(z)$, $CE(z)$, and $CU(z)$. Thus, prices of anarchy and satisfaction ratios are not defined in this case. However, since

$$CE(\mathbf{z}^*) = \min_{z \in \mathbb{R}^n} CE(z) = 0 = CU(\mathbf{z}^*) = \min_{z \in \mathbb{R}^n} CU(z),$$

the neutral consensus equilibrium is also socially optimal.

3.5 Oscillatory Behavior Of Opinions For 2 Agents

One can observe through numerical simulations that in certain cases, in a social network, we observe oscillatory or periodic behavior in the opinions of agents.

Analysing a general social network of n - agents is a separate research work and hence very hard. The paper sticks to analysing two-agent networks for simplicity and better visualization.

Understanding the analysis of two-agent networks is relevant since it can help us understand opinion behaviors or decision-making in important systems such as two-party politics, duopoly economic markets, etc.

The opinion dynamics for two agents is given by:

$$\begin{aligned}\dot{z}_1 &= S_1(z_1) + a_{12}(z_2 - z_1), \\ \dot{z}_2 &= S_2(z_2) - a_{21}(z_2 - z_1).\end{aligned}$$

For the two-agent dynamics, it can be easily verified that if the opinions $z_1(t)$ and $z_2(t)$ exhibit oscillatory behavior, then they will have the same fundamental period of oscillations.

Lemma 3.14 (Equal period of oscillations). *Suppose the opinions $z_1(t)$ and $z_2(t)$ exhibit oscillations with fundamental periods $T_1 > 0$ and $T_2 > 0$, respectively. Then, $T_1 = T_2$.*

Proof. One can rewrite the dynamics as:

$$\begin{aligned}\dot{z}_1 - S_1(z_1) + a_{12}z_1 &= a_{12}z_2, \\ \dot{z}_2 - S_2(z_2) + a_{21}z_2 &= a_{21}z_1.\end{aligned}$$

Now, suppose $z_1(t)$ and $z_2(t)$ are periodic with fundamental periods $T_1 > 0$ and $T_2 > 0$, respectively. Then, $S_1(z_1(t))$ and $\dot{z}_1$ (respectively, $S_2(z_2(t))$ and $\dot{z}_2$) are also periodic with fundamental period T_1 (respectively, T_2).

From the rewritten equations, it follows that the right-hand side of each equation must also be periodic. Therefore, both $z_1(t)$ and $z_2(t)$ must be periodic with a common period. Thus, there exist $m, n \in \mathbb{N}$ such that $T_1 = mT_2$ and $T_2 = nT_1$. Hence, $m = n = 1$, and we conclude that $T_1 = T_2$. $\square$

One can note that for periodic orbits, at least one agent should have an antagonistic influence on the other agent. Otherwise, as seen, we can guarantee convergence to a unique equilibrium. Also, both the influence weights should be non-zero as otherwise the socially closed agent i 's opinion would converge to m_i . The following result gives the necessary conditions for periodic solutions:

Lemma 3.15 (Necessary condition for existence of periodic solutions). *Opinions $z_1(t)$ and $z_2(t)$ exhibit periodic behavior only if all the following hold:*

- (i) $a_{12} < 0$ or $a_{21} < 0$,
- (ii) $a_{12}a_{21} \neq 0$,
- (iii) $(a_{21} + w_1 + w_2 + a_{12}) < 0$.

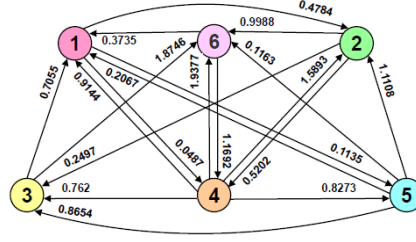
The first two conditions have been discussed above the lemma. One can prove the last condition using Bendixson's criterion and show that $(a_{21} + w_1 + w_2 + a_{12}) \geq 0$ implies a periodic orbit cannot exist.

The oscillatory behavior of opinions appears only when at least one of the agents has an antagonist influence on the other. This is similar to the boomerang effect, where at least one of the agents has an antagonist influence on another agent. As a result, the agent on whom there is an antagonist influence shifts his or her opinion away from the other agent. In such situations, the opinions of the agents could converge to a disagreement equilibrium or could possibly keep on oscillating forever, never converging to an equilibrium.

4 Simulations And Experiments

I have simulated most of the experiments in the base paper and have reproduced the results.

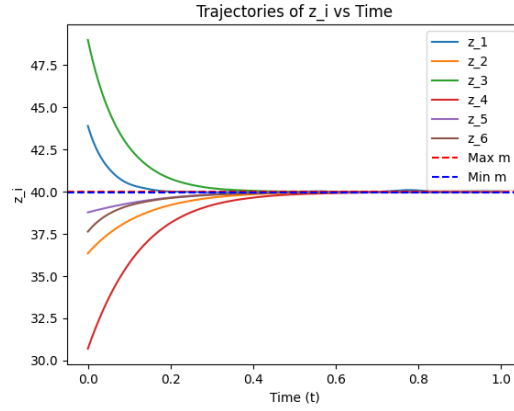
Consider the social network of 6 agents below. As seen, there are no antagonistic relations.



Assume the model dynamics has these parameters:

$$\begin{aligned} \mathbf{z}_0 &= [43.90, 36.34, 49.00, 30.69, 38.77, 37.63] \\ \mathbf{w} &= [1.09, 2.98, 2.59, 1.82, 1.01, 2.65] \\ \mathbf{r} &= [317.43, 814.54, 789.07, 852.26, 505.64, 635.66] \\ \mathbf{p} &= [225.22, 66.40, 71.33, 81.16, 165.39, 77.99] \end{aligned}$$

In this case $m_i = 40$ for all i and hence opinions converge to a consensus.



Calculating the deviations from internal preferences, $\mathbf{d}$, and dominance weights σ , one can verify the dominance in consensus:

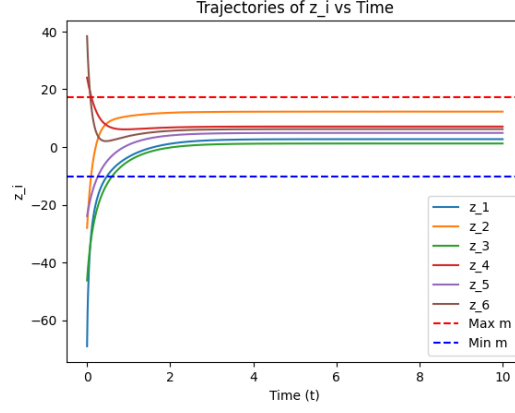
$$\mathbf{d} = [185.22, 26.40, 31.33, 41.16, 125.39, 37.99]$$

$$\sigma = [345.5, 2424.4, 2042.5, 1555.1, 510.4, 1684.8]$$

Next we consider a case when there is no consensus. Assume the same influence network.

$$\begin{aligned} \mathbf{z}_0 &= [-69.02, -28.03, -46.22, 24.01, -23.92, 38.38] \\ \mathbf{w} &= [0.22, 3.48, 0.56, 2.56, 0.95, 2.62] \\ \mathbf{r} &= [317.43, 814.54, 789.07, 852.26, 505.64, 635.66] \\ \mathbf{p} &= [-18.31, 18.92, -12.43, 6.68, 3.46, 7.00] \end{aligned}$$

The trajectories converge to a "disagreement equilibrium".



This is because the roots of the self-functions are not identical.

$$\mathbf{m} = [-8.74, 17.14, -10.10, 6.55, 3.38, 6.81]$$

Finally, we consider oscillatory trajectories of 2-agent opinion dynamics.

Consider a 2 agent social network with the following parameters:

$$\mathbf{z}_0 = [-0.0349 \quad -0.0039]$$

$$[a_{12} \quad a_{21}] = [-6.6667 \quad 3.3267]$$

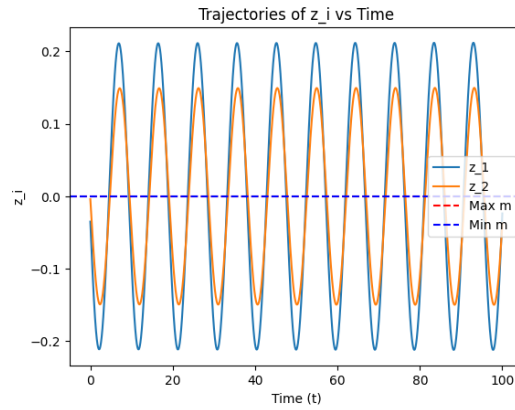
$$\mathbf{w} = [2 \quad 1.3333]$$

$$\mathbf{p} = [0 \quad 0]$$

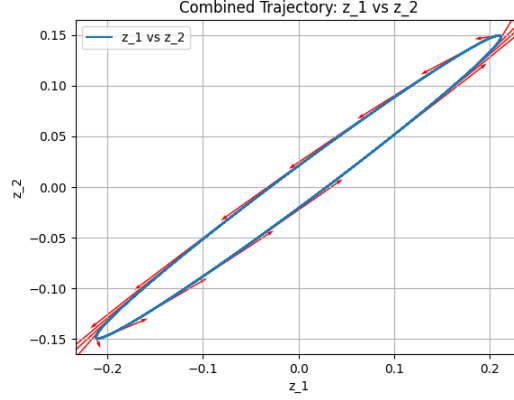
$$\mathbf{r} = [10 \quad 5]$$

Note that there is a chance that the system can have a neutral consensus with different initial conditions, since both agents have zero preference.

However, we observe oscillatory trajectories:



On the 2D plane one can observe the periodic orbit:



5 Conclusion And Future Work

Thus, a model of opinion dynamics under bounded resource constraints has been studied. Results regarding asymptotic behavior, namely convergence, ultimate boundedness and oscillatory behaviour of two-agent opinion dynamics under non-antagonistic influences. Consensus equilibria of the dynamics and Nash equilibria of the opinion formation game were studied. Dominance in consensus was interpreted as a measure of social power. The price of anarchy is a measure of how good a Nash equilibrium compared to the global optimum. A price of anarchy of unity is socially optimal. Experiments were carried out to reproduce theoretical results.

Further work could be carried out to explore the notion of influence maximization under resource constraints. Cooperation can be introduced in the influence maximization problem to improve the popularity or the opinion about an influencer. Opinion dynamics models can be used in multi-agent coordination games, where agents choose a strategy based on an opinion vector that evolves with time,

Finally, the model has been extended to multiple topics where each agent holds a vector of opinions, one for each topic. The vector could be composed of different choices, upon which each agent has his or her personal opinion. I have shown that the modified model asymptotically also results in an ultimately bounded opinion profile. One can extend the analysis of this model and the equilibria of the modified dynamics in future work.

Additionally, one might not know one's internal preferences. In such a case, sometimes an agent may want to partly optimize one's self utility function, which might be computationally expensive, hence partly following the opinion of the crowd. I have modeled the dynamics where each agent's resources are limited and her opinion evolves due to a linear combination of both factors.

Acknowledgments

I would like to thank Prof. Pavankumar Tallapragada for his advice and guidance in this paper.

I have used AI for writing LaTeX code.

Appendix A: Extending the model to multiple topics

One can easily extend the above dynamics to K —topics, where each agent has an opinion and an internal preference on each topic, and governed by the following dynamics:

$$\dot{z}_{ik} = -w_i(z_{ik} - p_{ik}) + \sum_{j \in V} a_{ij}(z_{jk} - z_{ik}) - \frac{z_{ik}^3}{r_i}$$

However we don't see the coupling terms, that is z_{ik} can grow unbounded relative to z_{il} where k, l are 2 topics. Hence, we need a modified utility function, to update the opinion of each (agent,topic) pair.

Define the utility of agent i as the following:

$$U_i(\mathbf{z}_i, \mathbf{z}_{-i}) = -\frac{w_i}{2}(z_{ik} - p_{ik})^2 - \sum_{j \in V} \frac{a_{ij}}{2}(z_{jk} - z_{ik})^2 - \frac{\|\mathbf{z}_i\|^4}{4r_i}$$

where $\mathbf{z}_i$ represents the K dimensional vector of opinions of agent i .

One can take the partial derivatives of the above utility function with respect to the opinions to derive the dynamics of the opinion for each (agent,topic) pair.

$$\dot{z}_{ik} = -w_i(z_{ik} - p_{ik}) + \sum_{j \in V} a_{ij}(z_{jk} - z_{ik}) - \frac{(z_{ik}^3 + 0.5 \sum_{l \in [K], l \neq k} z_{ik} z_{il}^2)}{r_i}$$

Lemma 5.1. *Under these dynamics, opinions are ultimately bounded, just like the single topic case(Theorem 3.4). One can choose the Lyapunov function $V = 0.5\mathbf{z}^T \mathbf{z}$, where $\mathbf{z}$ is a flattened vector of all agents' opinions to similarly show ultimate boundedness.*

Experiments

I took a 2-agent 2-topic social network dynamics and simulated the trajectories. Consider the following parameters of the model:

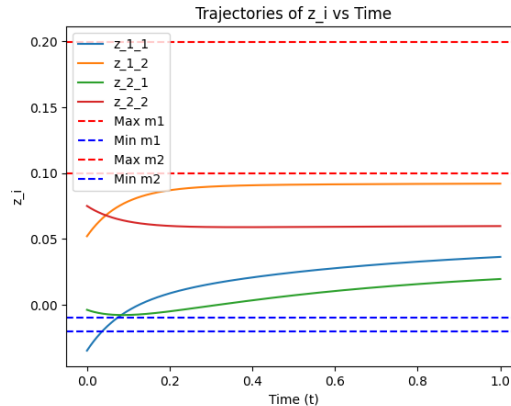
$$\begin{aligned} \mathbf{z}_0 &= \begin{bmatrix} -0.0349 & 0.052 \\ -0.0039 & 0.075 \end{bmatrix} \\ A &= \begin{bmatrix} 0 & 6.6667 \\ 3.3267 & 0 \end{bmatrix} \\ \mathbf{w} &= [2 \quad 1.3333] \\ \mathbf{p} &= \begin{bmatrix} 0.1 & 0.2 \\ -0.01 & -0.02 \end{bmatrix} \\ \mathbf{r} &= [10 \quad 5] \end{aligned}$$

Each row of $\mathbf{z}_0$ and $\mathbf{p}$ is a vector of initial opinions and internal preferences of an agent respectively.

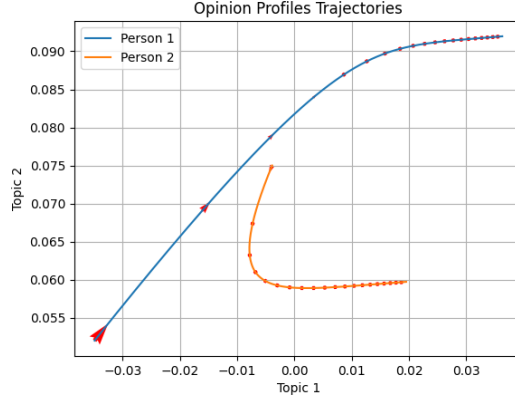
I calculated the roots of the self portion of the "wrong" dynamics, i.e the one without coupling and obtained the following matrix $\mathbf{m}$:

$$\mathbf{m} = \begin{bmatrix} 0.09995007 & -0.00999985 \\ 0.19960238 & -0.0199988 \end{bmatrix}$$

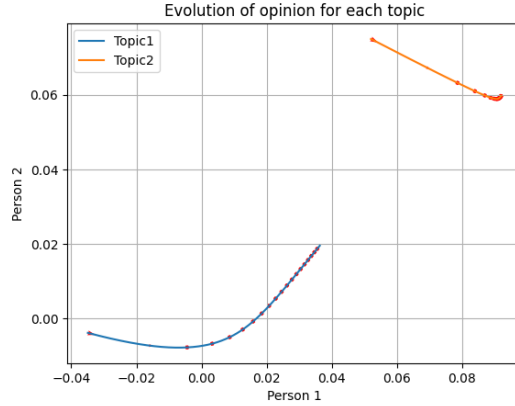
Plotting the trajectories, we see that the opinions for each topic converge inside $\mathcal{M} = [m_{min}, m_{max}]$ for that topic. However, this is not always true.



For each person(agent), I plotted the trajectory of opinions of topics 1 and 2 on a 2D graph:



And also the trajectory of the opinions of both people fixing a topic:



Appendix B: When computing self preference is computationally expensive

Sometimes, people are not aware of their true internal preference on a topic. They might even have multiple internal preferences, where the opinion attains equilibrium. Thus the self part of the utility function can have multiple maximizers. More over computing the gradient of this self part can be computationally expensive.

Not all agents have equal amount of resources, enough to compute this gradient.

Thus, the dynamics of each agent's opinion are constrained by the amount of resources the agent has.

We model these constraints by a dynamics where each agent updates her opinion by the gradient of the self function sometimes, and rest of the times, she follows the crowd.

$$\dot{z}_i = f(z_i)a_i + b_i\left(\sum_{k \in V} a_{ik}(z_k - z_i)\right)$$

Here f is the gradient function, a_i and b_i are non-negative parameters that highlight the resource constraints of agent i .

For each agent, I take

$$f(z_i) = - \prod_{k \in [n_i]} (x - p_{ik})$$

where each agent i has n_i maximizers of their self-utility function p_{ik}

I have conducted simulations on a two-agent social network, the first case where the agents converge to a consensus, and for other initial conditions, we even see unbounded opinions. Note that the later was not possible in the vanilla model of the paper.

Consensus Case

Consider two agent single opinion dynamics with the following parameters. Here A represents the influence matrix. P is a nested list where each element is a vector of the maximizers of the self-function of the agent.

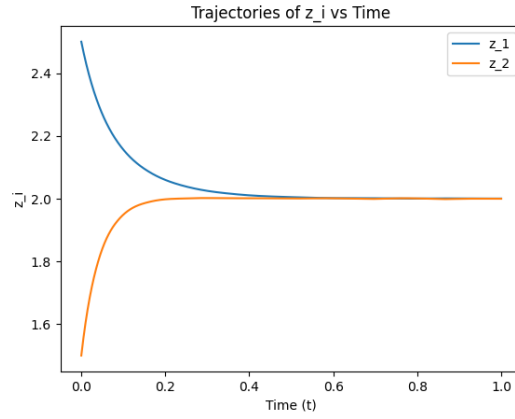
$$\mathbf{z}_0 = [2.5 \quad 1.5]$$

$$A = \begin{bmatrix} 0 & 6.6667 \\ 3.3267 & 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 2 & 5 & 6 \\ -1 & 2 & 4 & 7 \end{bmatrix}$$

$$\mathbf{a} = [0.4 \quad 0.6] \quad \mathbf{b} = [0.6 \quad 0.4]$$

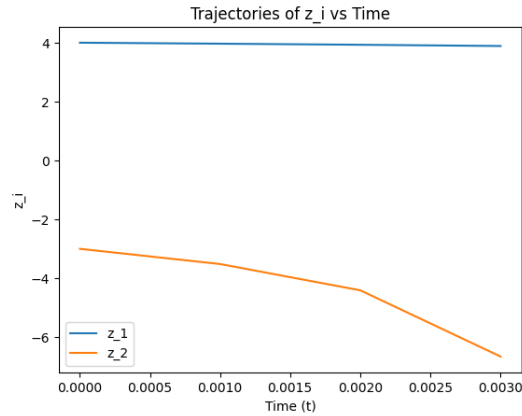
The trajectories reach a consensus equilibrium as can be seen:



Unbounded opinions

$$\mathbf{z}_0 = [4.0 \quad -3.0]$$

In this case we see that an agent's opinion converges, while the other agent's opinion diverges.



References

1. Wankhede et al., "Opinion Dynamics for Utility Maximizing Agents: Exploring the Impact of the Resource Penalty," IEEE Transactions on Control of Network Systems, 2024.

2. J. R. French Jr, "A formal theory of social power," *Psychological Review*, vol. 63, no. 3, p. 181, 1956.
3. M. H. DeGroot, "Reaching a consensus," *Journal of the American Statistical Association*, vol. 69, no. 345, pp. 118–121, 1974.
4. R. P. Abelson, "Mathematical models of the distribution of attitudes under controversy," *Contributions to Mathematical Psychology*, 1964.
5. M. Taylor, "Towards a mathematical theory of influence and attitude change," *Human Relations*, vol. 21, no. 2, pp. 121–139, 1968.
6. N. E. Friedkin and E. C. Johnsen, "Social influence and opinions," *Journal of Mathematical Sociology*, vol. 15, no. 3-4, pp. 193–206, 1990.
7. R. Hegselmann and U. Krause, "Opinion dynamics and bounded confidence models, analysis, and simulation," *Journal of Artificial Societies and Social Simulation*, vol. 5, no. 3, 2002.
8. C. Altafini, "Consensus problems on networks with antagonistic interactions," *IEEE Transactions on Automatic Control*, vol. 58, no. 4, pp. 935–946, 2013.
9. G. Deffuant, D. Neau, F. Amblard, and G. Weisbuch, "Mixing beliefs among interacting agents," *Advances in Complex Systems*, vol. 3, no. 01n04, pp. 87–98, 2000.
10. A. V. Proskurnikov and R. Tempo, "A tutorial on modeling and analysis of dynamic social networks. Part I," **Annual Reviews in Control**, vol. 43, pp. 65–79, 2017.
11. ———, "A tutorial on modeling and analysis of dynamic social networks. Part II," **Annual Reviews in Control**, vol. 45, pp. 166–190, 2018.
12. B. D. Anderson, F. Dabbene, A. V. Proskurnikov, C. Ravazzi, and M. Ye, "Dynamical networks of social influence: Modern trends and perspectives," **IFAC-PapersOnLine**, vol. 53, no. 2, pp. 17616–17627, 2020.
13. Q. Zha, G. Kou, H. Zhang, H. Liang, X. Chen, C.-C. Li, and Y. Dong, "Opinion dynamics in finance and business: a literature review and research opportunities," **Financial Innovation**, vol. 6, pp. 1–22, 2020.
14. Z. Meng, G. Shi, K. H. Johansson, M. Cao, and Y. Hong, "Behaviors of networks with antagonistic interactions and switching topologies," **Automatica**, vol. 73, pp. 110–116, 2016.
15. J. Liu, X. Chen, T. Başar, and M. A. Belabbas, "Exponential convergence of the discrete-and continuous-time Altafini models," **IEEE Transactions on Automatic Control**, vol. 62, no. 12, pp. 6168–6182, 2017.
16. P. Cisneros-Velarde, K. S. Chan, and F. Bullo, "Polarization and fluctuations in signed social networks," **IEEE Transactions on Automatic Control**, vol. 66, no. 8, pp. 3789–3793, 2021.
17. M. Ye, Y. Qin, A. Govaert, B. D. Anderson, and M. Cao, "An influence network model to study discrepancies in expressed and private opinions," **Automatica**, vol. 107, pp. 371–381, 2019.
18. P. Jia, A. Mirtabatabaei, N. E. Friedkin, and F. Bullo, "Opinion dynamics and the evolution of social power in influence networks," **SIAM Review**, vol. 57, no. 3, pp. 367–397, 2015.
19. M. Ye and B. D. Anderson, "Modelling of individual behaviour in the DeGroot–Friedkin self-appraisal dynamics on social networks," in **2019 18th European Control Conference (ECC)**, 2019, pp. 2011–2017.
20. Y. Tian, P. Jia, A. MirTabatabaei, L. Wang, N. E. Friedkin, and F. Bullo, "Social power evolution in influence networks with stubborn individuals," **IEEE Transactions on Automatic Control**, vol. 67, no. 2, pp. 574–588, 2022.
21. A. Bizyaeva, A. Franci, and N. E. Leonard, "Sustained oscillations in multi-topic belief dynamics over signed networks," in **2023 American Control Conference (ACC)**, 2023, pp. 4296–4301.
22. J. R. Marden, G. Arslan, and J. S. Shamma, "Cooperative control and potential games," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 39, no. 6, pp. 1393–1407, 2009.

23. J. Ghaderi and R. Srikant, "Opinion dynamics in social networks with stubborn agents: Equilibrium and convergence rate," **Automatica**, vol. 50, no. 12, pp. 3209–3215, 2014.
24. D. Bindel, J. Kleinberg, and S. Oren, "How bad is forming your own opinion?" **Games and Economic Behavior**, vol. 92, pp. 248–265, 2015.
25. P. Groeber, J. Lorenz, and F. Schweitzer, "Dissonance minimization as a microfoundation of social influence in models of opinion formation," **The Journal of Mathematical Sociology**, vol. 38, no. 3, pp. 147–174, 2014.
26. S. R. Etesami, "Open-loop equilibrium strategies for dynamic influence maximization game over social networks," **IEEE Control Systems Letters**, vol. 6, pp. 1496–1500, 2022.
27. S. R. Etesami and T. Başar, "Game-theoretic analysis of the Hegselmann-Krause model for opinion dynamics in finite dimensions," **IEEE Transactions on Automatic Control**, vol. 60, no. 7, pp. 1886–1897, 2015.
28. L. Zino, M. Ye, and M. Cao, "A two-layer model for coevolving opinion dynamics and collective decision-making in complex social systems," **Chaos: An Interdisciplinary Journal of Nonlinear Science**, vol. 30, no. 8, p. 083107, 2020.
29. H. D. Aghbolagh, M. Ye, L. Zino, Z. Chen, and M. Cao, "Coevolutionary dynamics of actions and opinions in social networks," **IEEE Transactions on Automatic Control**, vol. 68, no. 12, pp. 7708–7723, 2023.
30. S. Park, A. Bizyaeva, M. Kawakatsu, A. Franci, and N. E. Leonard, "Tuning cooperative behavior in games with nonlinear opinion dynamics," **IEEE Control Systems Letters**, vol. 6, pp. 2030–2035, 2022.
31. S. R. Etesami, S. Bolouki, A. Nedić, T. Başar, and H. V. Poor, "Influence of conformist and manipulative behaviors on public opinion," **IEEE Transactions on Control of Network Systems**, vol. 6, no. 1, pp. 202–214, 2019.
32. H. K. Khalil, **Nonlinear Systems**, 3rd ed. Upper Saddle River, NJ: Prentice-Hall, 2002.
33. J. Guckenheimer and P. Holmes, **Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields**. Springer, 2013.